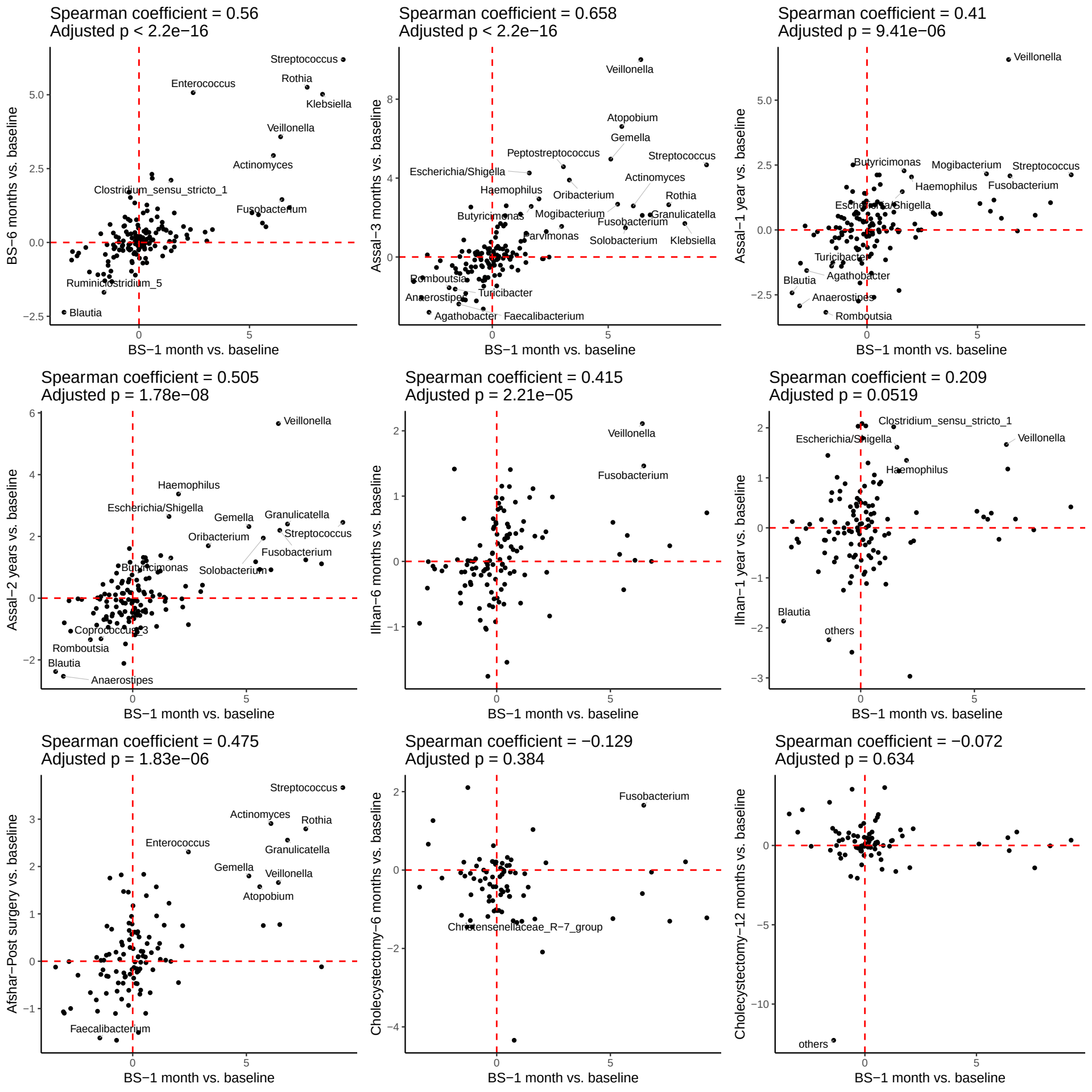
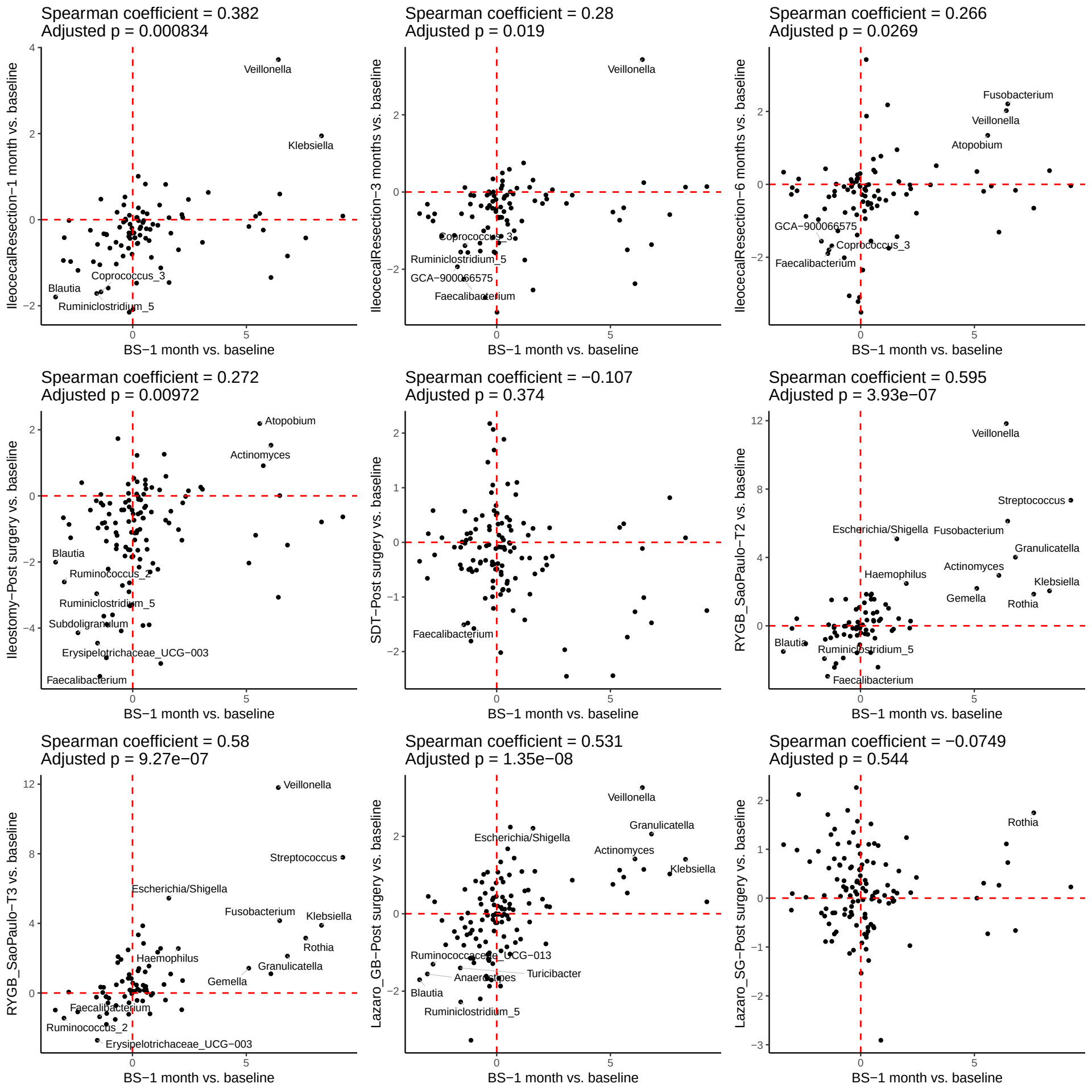
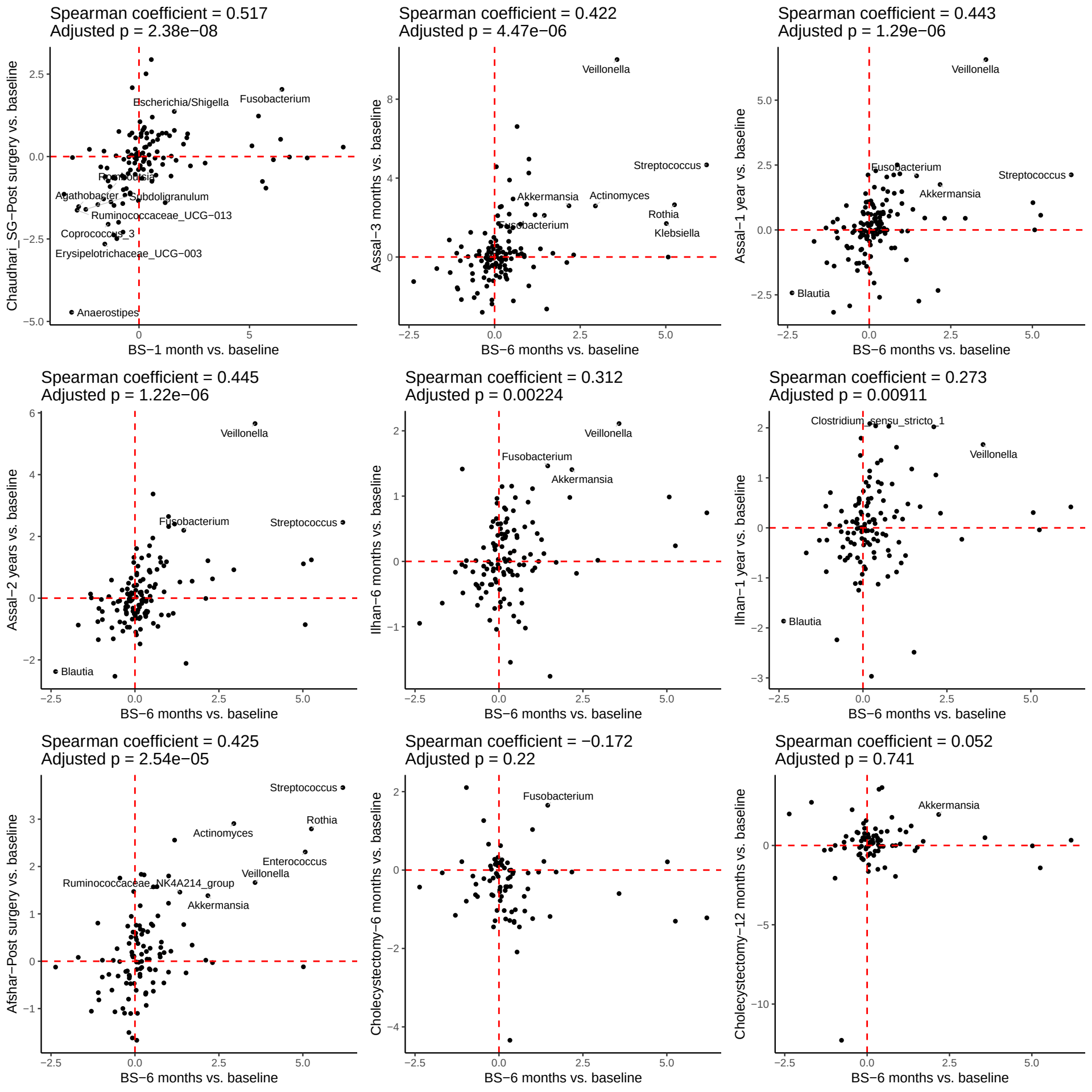
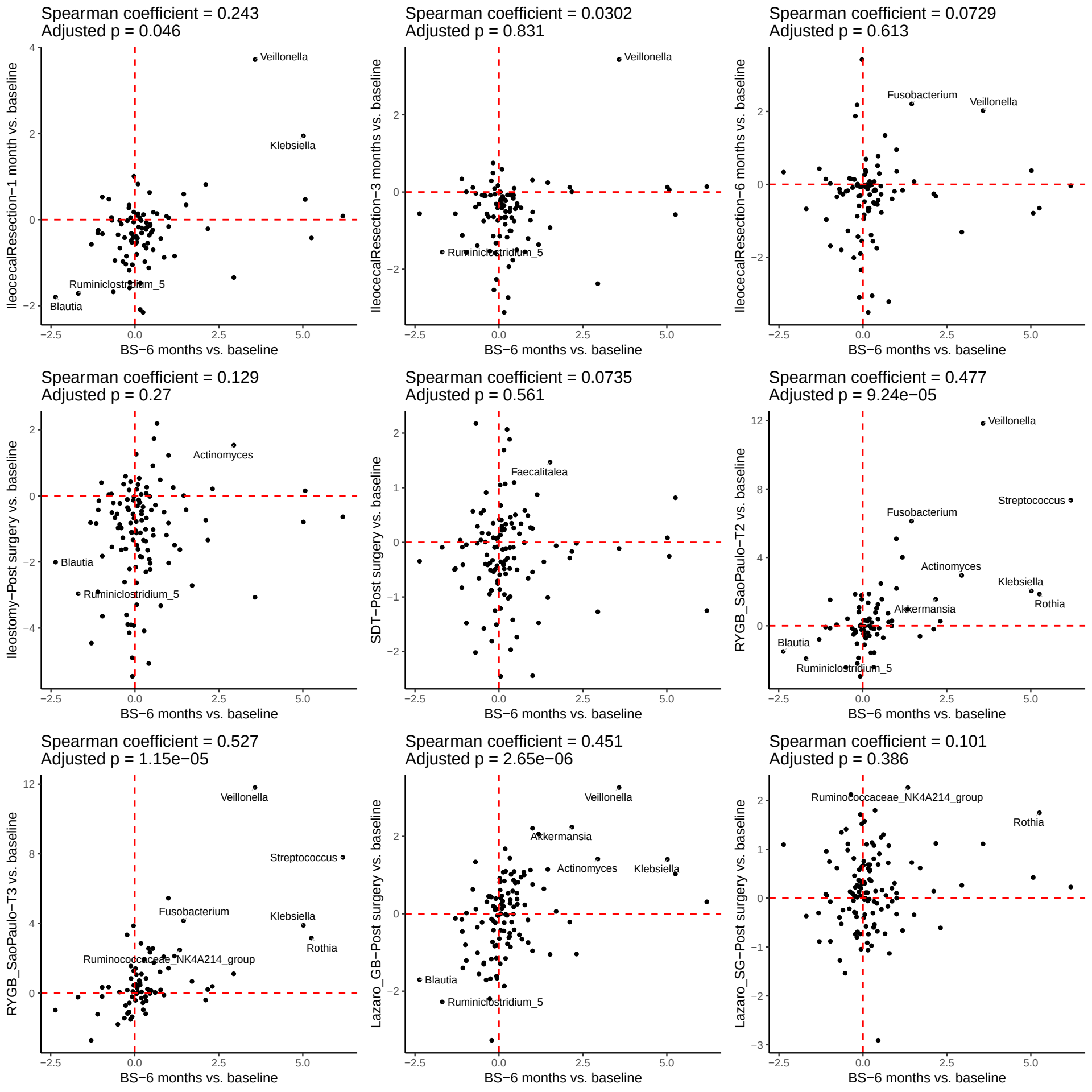
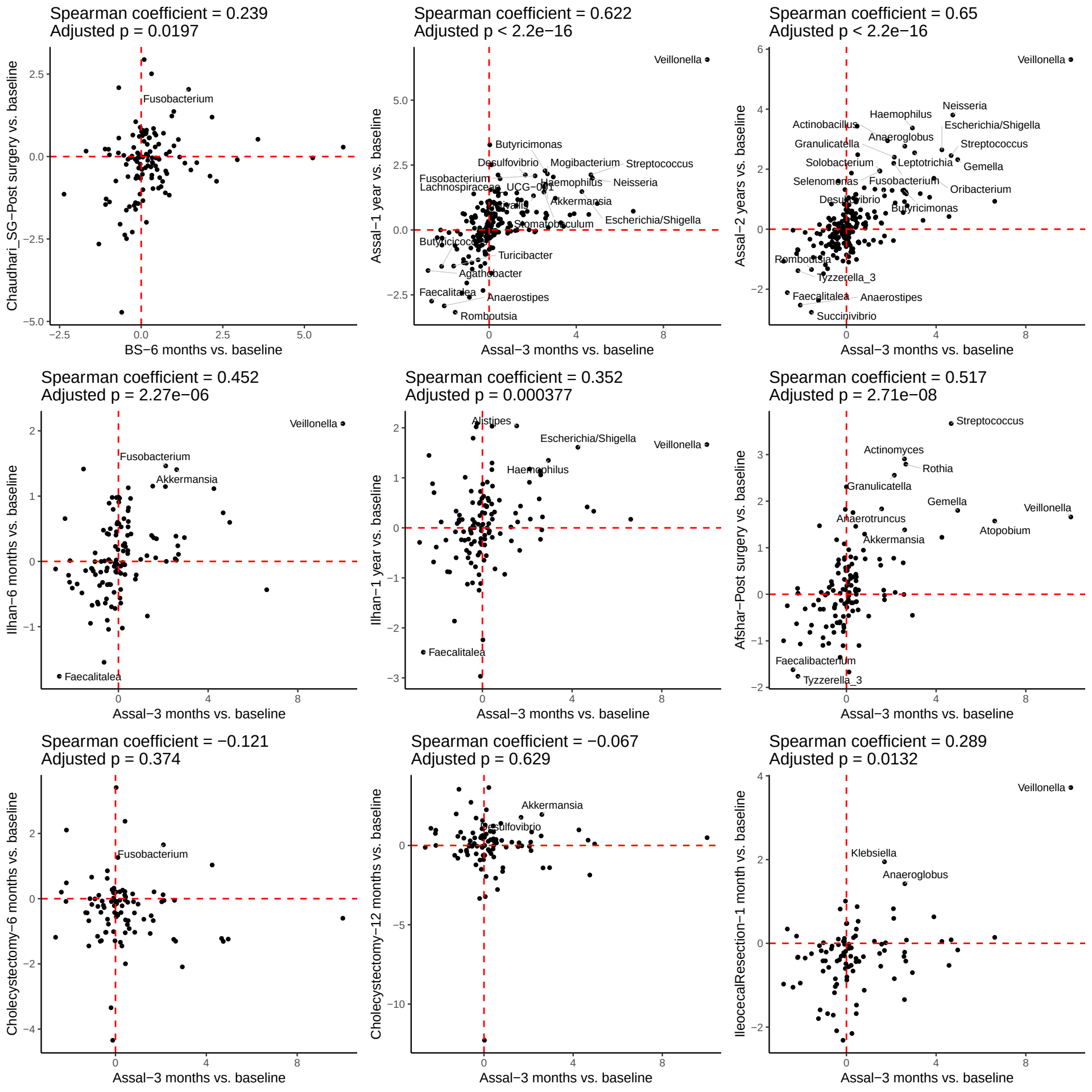
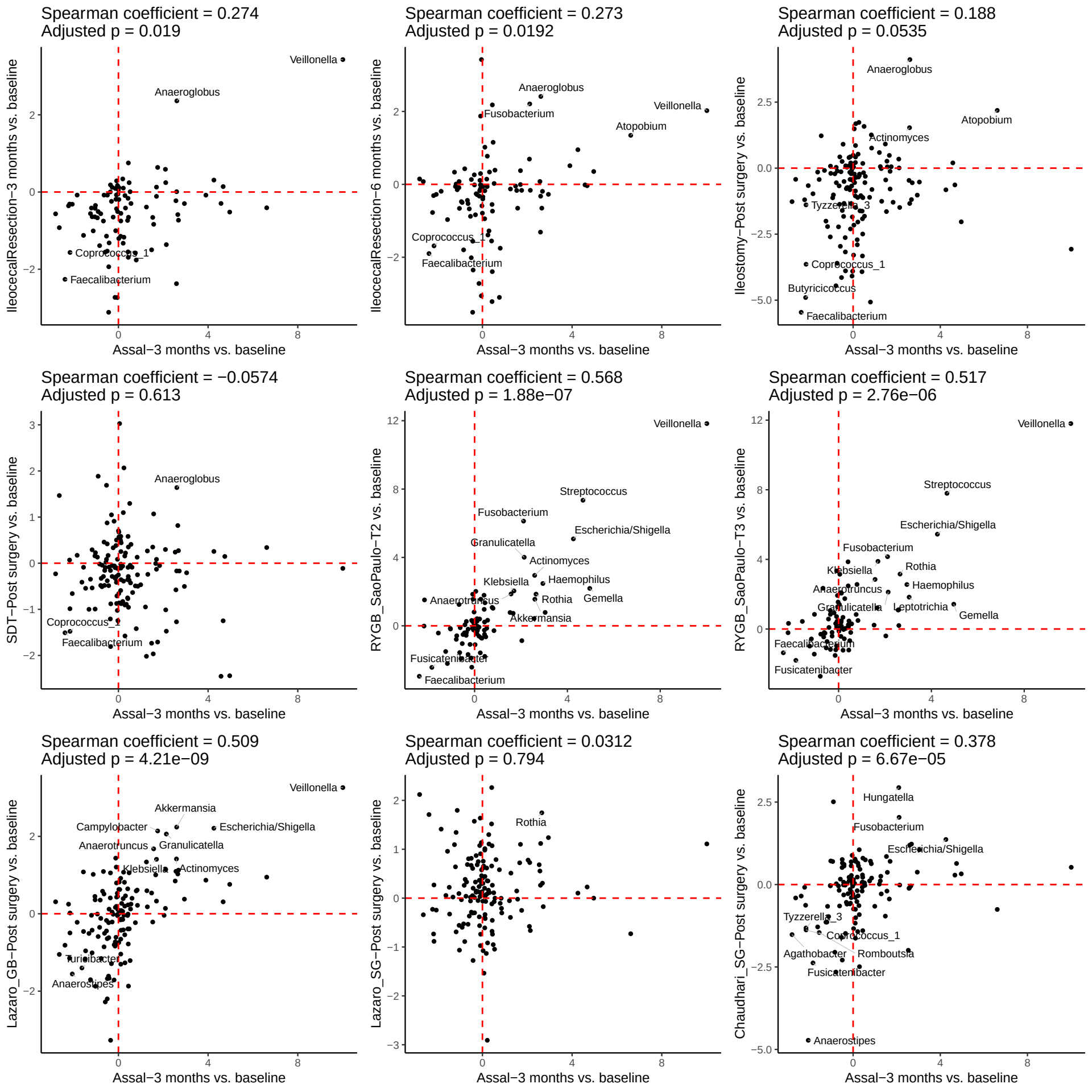

Spearman coefficient = 0.56
Adjusted p < 2.2e−16
BS−6 months vs. baseline
BS−1 month vs. baseline
Streptococcus
Rothia
Enterococcus
Klebsiella
Veillonella
Actinomyces
Clostridium_sensu_stricto_1
Fusobacterium
Ruminiclostridium_5
Blautia
Spearman coefficient = 0.658
Adjusted p < 2.2e−16
Assal−3 months vs. baseline
BS−1 month vs. baseline
Veillonella
Atopobium
Gemella
Peptostreptococcus
Streptococcus
Escherichia/Shigella
Actinomyces
Haemophilus
Oribacterium
Rothia
Butyricimonas
Mogibacterium
Granulicatella
Fusobacterium
Parvimonas
Solobacterium
Klebsiella
Romboutsia
Turicibacter
Anaerostipes
Agathobacter
Faecalibacterium
Spearman coefficient = 0.41
Adjusted p = 9.41e−06
Assal−1 year vs. baseline
BS−1 month vs. baseline
Veillonella
Butyricimonas
Mogibacterium
Streptococcus
Haemophilus
Fusobacterium
Escherichia/Shigella
Turicibacter
Agathobacter
Blautia
Anaerostipes
Romboutsia
Spearman coefficient = 0.505
Adjusted p = 1.78e−08
Assal−2 years vs. baseline
BS−1 month vs. baseline
Veillonella
Haemophilus
Escherichia/Shigella
Gemella
Granulicatella
Oribacterium
Streptococcus
Fusobacterium
Butyricimonas
Solobacterium
Coprococcus_3
Romboutsia
Blautia
Anaerostipes
Spearman coefficient = 0.415
Adjusted p = 2.21e−05
Ilhan−6 months vs. baseline
BS−1 month vs. baseline
Veillonella
Fusobacterium
Spearman coefficient = 0.209
Adjusted p = 0.0519
Ilhan−1 year vs. baseline
BS−1 month vs. baseline
Clostridium_sensu_stricto_1
Escherichia/Shigella
Veillonella
Haemophilus
Blautia
others
Spearman coefficient = 0.475
Adjusted p = 1.83e−06
Afshar−Post surgery vs. baseline
BS−1 month vs. baseline
Streptococcus
Actinomyces
Rothia
Enterococcus
Granulicatella
Gemella
Veillonella
Atopobium
Faecalibacterium
Spearman coefficient = −0.129
Adjusted p = 0.384
Cholecystectomy−6 months vs. baseline
BS−1 month vs. baseline
Fusobacterium
Christensenellaceae_R−7_group
Spearman coefficient = −0.072
Adjusted p = 0.634
Cholecystectomy−12 months vs. baseline
BS−1 month vs. baseline
others

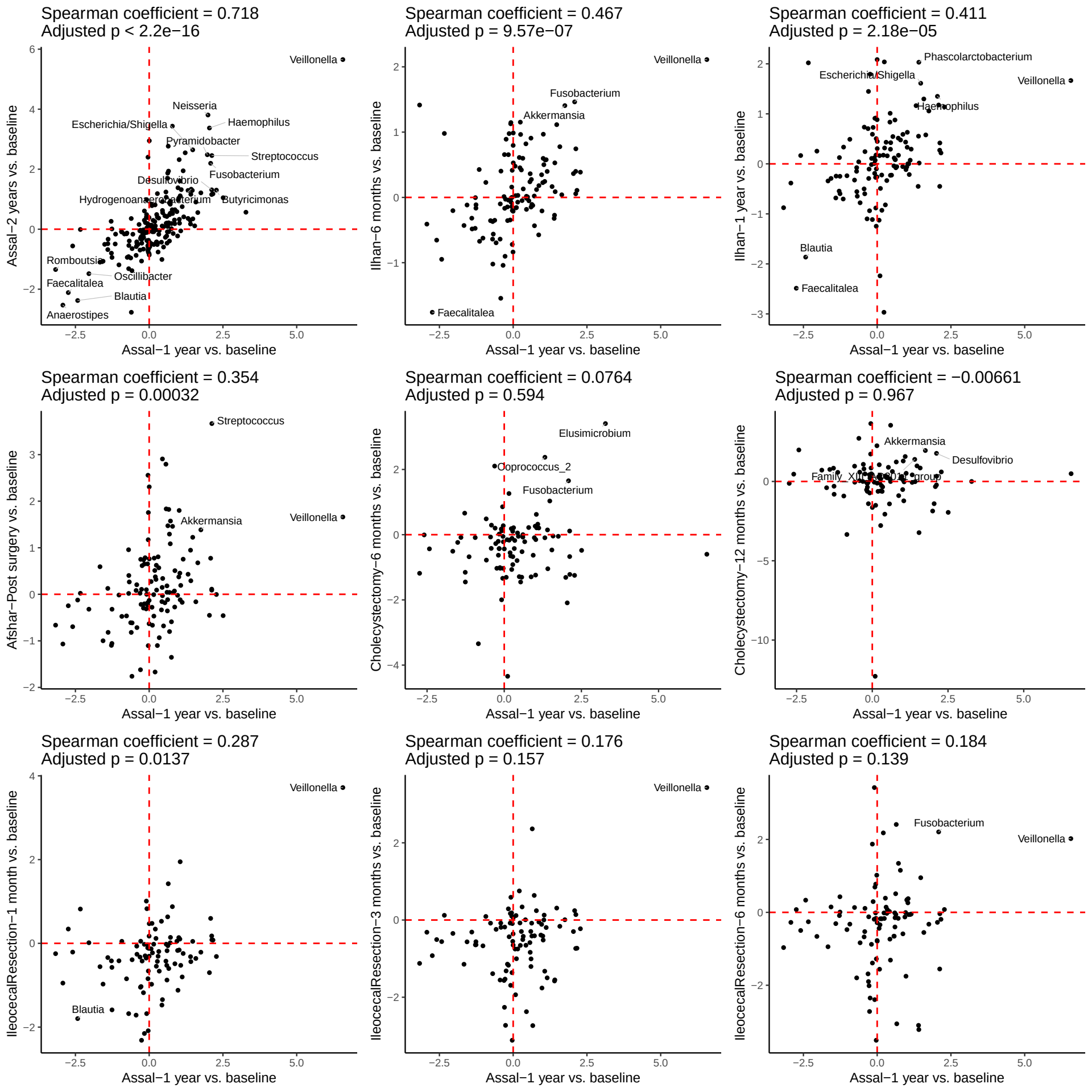

Spearman coefficient = 0.382
Adjusted p = 0.000834
IleocecalResection−1 month vs. baseline
BS−1 month vs. baseline
Veillonella
Klebsiella
Coprococcus_3
Blautia
Ruminiclostridium_5
Spearman coefficient = 0.28
Adjusted p = 0.019
IleocecalResection−3 months vs. baseline
BS−1 month vs. baseline
Veillonella
Coprococcus_3
Ruminiclostridium_5
GCA−900066575
Faecalibacterium
Spearman coefficient = 0.266
Adjusted p = 0.0269
IleocecalResection−6 months vs. baseline
BS−1 month vs. baseline
Fusobacterium
Veillonella
Atopobium
GCA−900066575
Coprococcus_3
Faecalibacterium
Spearman coefficient = 0.272
Adjusted p = 0.00972
Ileostomy−Post surgery vs. baseline
BS−1 month vs. baseline
Atopobium
Actinomyces
Blautia
Ruminococcus_2
Ruminiclostridium_5
Subdoligranulum
Erysipelotrichaceae_UCG−003
Faecalibacterium
Spearman coefficient = −0.107
Adjusted p = 0.374
SDT−Post surgery vs. baseline
BS−1 month vs. baseline
Faecalibacterium
Spearman coefficient = 0.595
Adjusted p = 3.93e−07
RYGB_SaoPaulo−T2 vs. baseline
BS−1 month vs. baseline
Veillonella
Streptococcus
Escherichia/Shigella
Fusobacterium
Granulicatella
Actinomyces
Haemophilus
Klebsiella
Gemella
Rothia
Blautia
Ruminiclostridium_5
Faecalibacterium
Spearman coefficient = 0.58
Adjusted p = 9.27e−07
RYGB_SaoPaulo−T3 vs. baseline
BS−1 month vs. baseline
Veillonella
Streptococcus
Escherichia/Shigella
Fusobacterium
Klebsiella
Rothia
Haemophilus
Granulicatella
Gemella
Faecalibacterium
Ruminococcus_2
Erysipelotrichaceae_UCG−003
Spearman coefficient = 0.531
Adjusted p = 1.35e−08
Lazaro_GB−Post surgery vs. baseline
BS−1 month vs. baseline
Veillonella
Granulicatella
Escherichia/Shigella
Actinomyces
Klebsiella
Ruminococcaceae_UCG−013
Turicibacter
Anaerostipes
Blautia
Ruminiclostridium_5
Spearman coefficient = −0.0749
Adjusted p = 0.544
Lazaro_SG−Post surgery vs. baseline
BS−1 month vs. baseline
Rothia

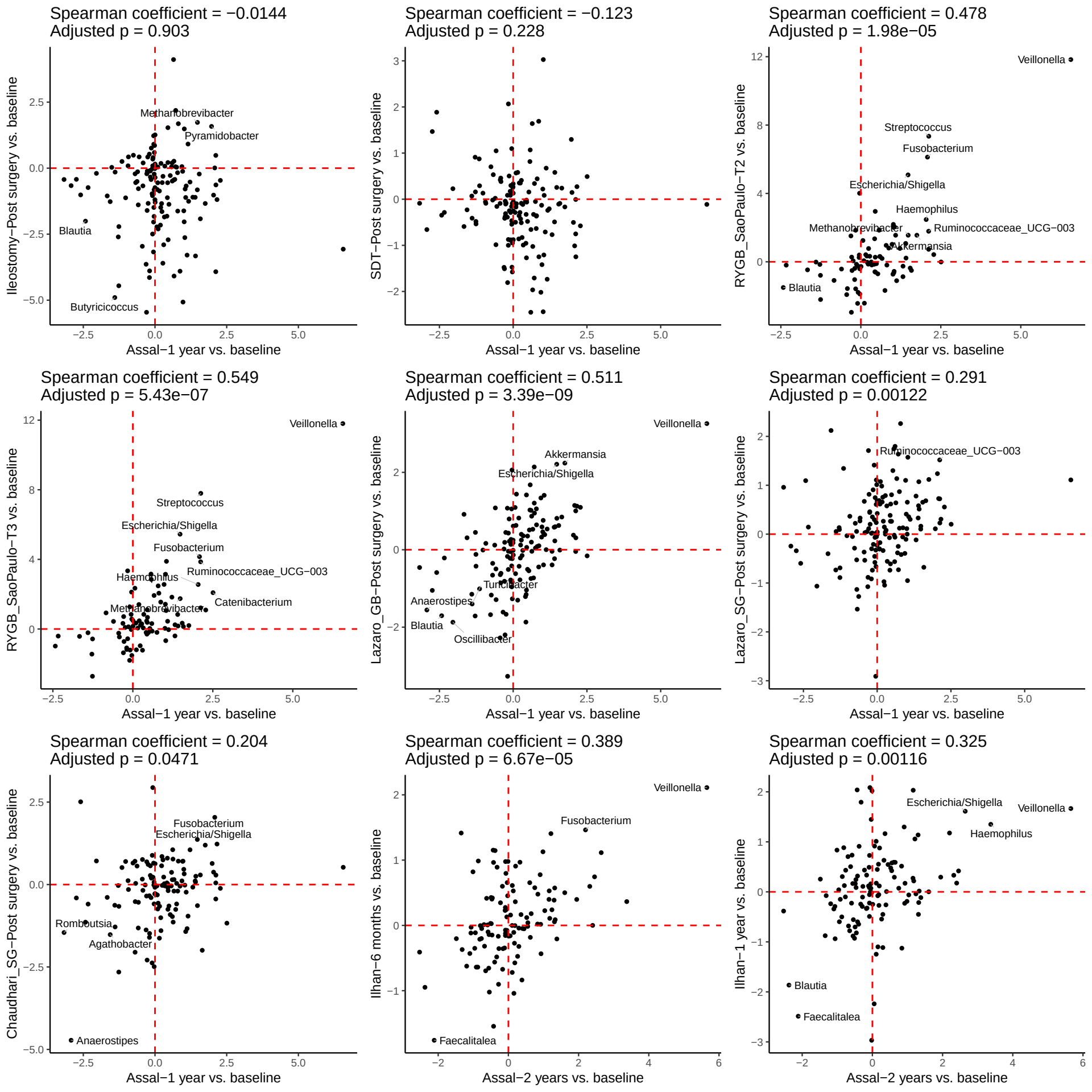

Spearman coefficient = 0.517
Adjusted p = 2.38e−08
Chaudhari_SG−Post surgery vs. baseline
BS−1 month vs. baseline
Escherichia/Shigella
Fusobacterium
Roseburia
Agathobacter
Subdoligranulum
Ruminococcaceae_UCG−013
Coprococcus_3
Erysipelotrichaceae_UCG−003
Anaerostipes
Spearman coefficient = 0.422
Adjusted p = 4.47e−06
Assal−3 months vs. baseline
BS−6 months vs. baseline
Veillonella
Streptococcus
Akkermansia
Actinomyces
Rothia
Fusobacterium
Klebsiella
Spearman coefficient = 0.443
Adjusted p = 1.29e−06
Assal−1 year vs. baseline
BS−6 months vs. baseline
Veillonella
Fusobacterium
Streptococcus
Akkermansia
Blautia
Spearman coefficient = 0.445
Adjusted p = 1.22e−06
Assal−2 years vs. baseline
BS−6 months vs. baseline
Veillonella
Fusobacterium
Streptococcus
Blautia
Spearman coefficient = 0.312
Adjusted p = 0.00224
Ilhan−6 months vs. baseline
BS−6 months vs. baseline
Veillonella
Fusobacterium
Akkermansia
Spearman coefficient = 0.273
Adjusted p = 0.00911
Ilhan−1 year vs. baseline
BS−6 months vs. baseline
Clostridium_sensu_stricto_1
Veillonella
Blautia
Spearman coefficient = 0.425
Adjusted p = 2.54e−05
Afshar−Post surgery vs. baseline
BS−6 months vs. baseline
Streptococcus
Rothia
Actinomyces
Enterococcus
Veillonella
Ruminococcaceae_NK4A214_group
Akkermansia
Spearman coefficient = −0.172
Adjusted p = 0.22
Cholecystectomy−6 months vs. baseline
BS−6 months vs. baseline
Fusobacterium
Spearman coefficient = 0.052
Adjusted p = 0.741
Cholecystectomy−12 months vs. baseline
BS−6 months vs. baseline
Akkermansia

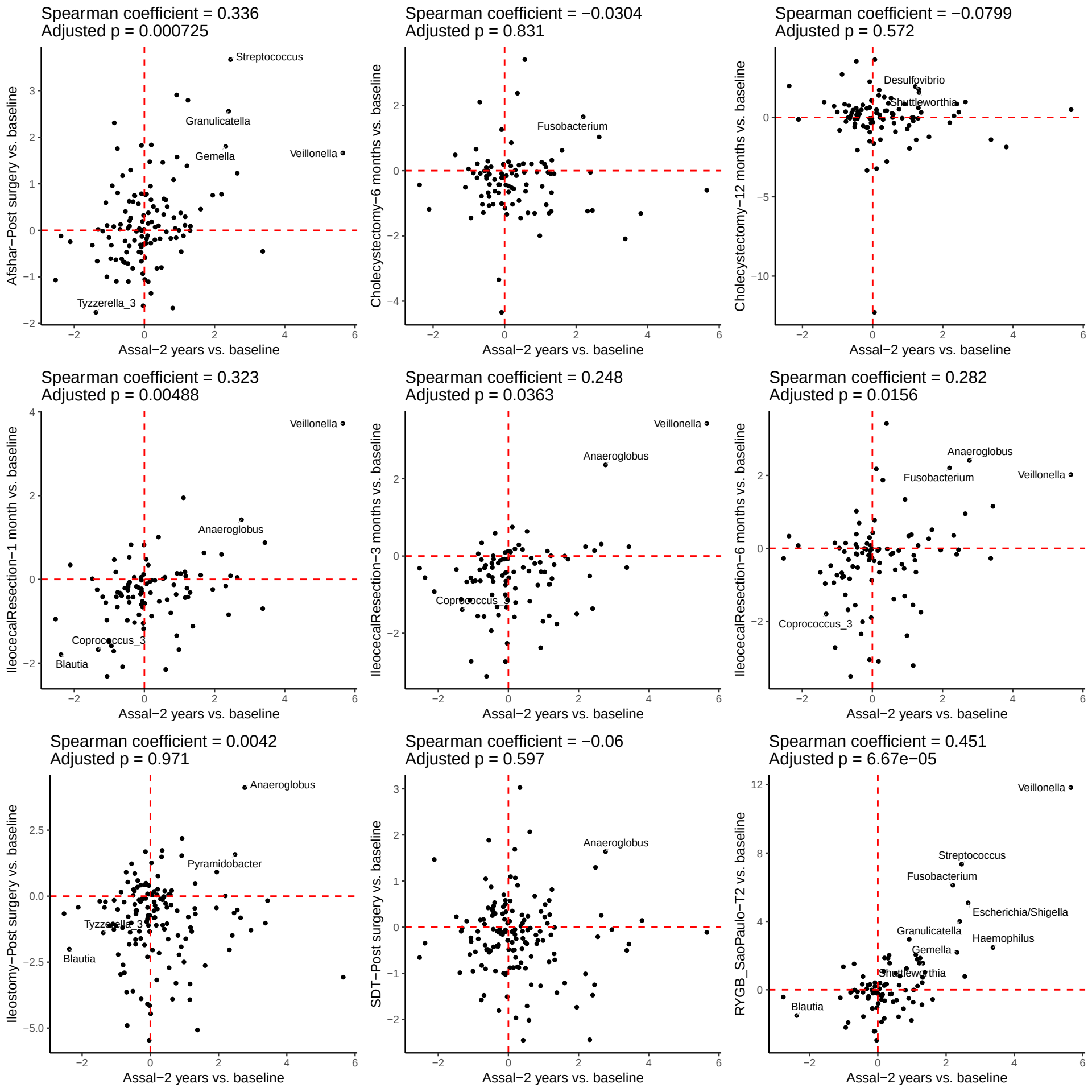

Spearman coefficient = 0.243
Adjusted p = 0.046
IleocecalResection−1 month vs. baseline
BS−6 months vs. baseline
Veillonella
Klebsiella
Ruminiclostridium_5
Blautia
Spearman coefficient = 0.0302
Adjusted p = 0.831
IleocecalResection−3 months vs. baseline
BS−6 months vs. baseline
Veillonella
Ruminiclostridium_5
Spearman coefficient = 0.0729
Adjusted p = 0.613
IleocecalResection−6 months vs. baseline
BS−6 months vs. baseline
Fusobacterium
Veillonella
Spearman coefficient = 0.129
Adjusted p = 0.27
Ileostomy−Post surgery vs. baseline
BS−6 months vs. baseline
Actinomyces
Blautia
Ruminiclostridium_5
Spearman coefficient = 0.0735
Adjusted p = 0.561
SDT−Post surgery vs. baseline
BS−6 months vs. baseline
Faecalitalea
Spearman coefficient = 0.477
Adjusted p = 9.24e−05
RYGB_SaoPaulo−T2 vs. baseline
BS−6 months vs. baseline
Veillonella
Streptococcus
Fusobacterium
Actinomyces
Klebsiella
Rothia
Akkermansia
Blautia
Ruminiclostridium_5
Spearman coefficient = 0.527
Adjusted p = 1.15e−05
RYGB_SaoPaulo−T3 vs. baseline
BS−6 months vs. baseline
Veillonella
Streptococcus
Fusobacterium
Klebsiella
Rothia
Ruminococcaceae_NK4A214_group
Spearman coefficient = 0.451
Adjusted p = 2.65e−06
Lazaro_GB−Post surgery vs. baseline
BS−6 months vs. baseline
Veillonella
Akkermansia
Actinomyces
Klebsiella
Blautia
Ruminiclostridium_5
Spearman coefficient = 0.101
Adjusted p = 0.386
Lazaro_SG−Post surgery vs. baseline
BS−6 months vs. baseline
Ruminococcaceae_NK4A214_group
Rothia

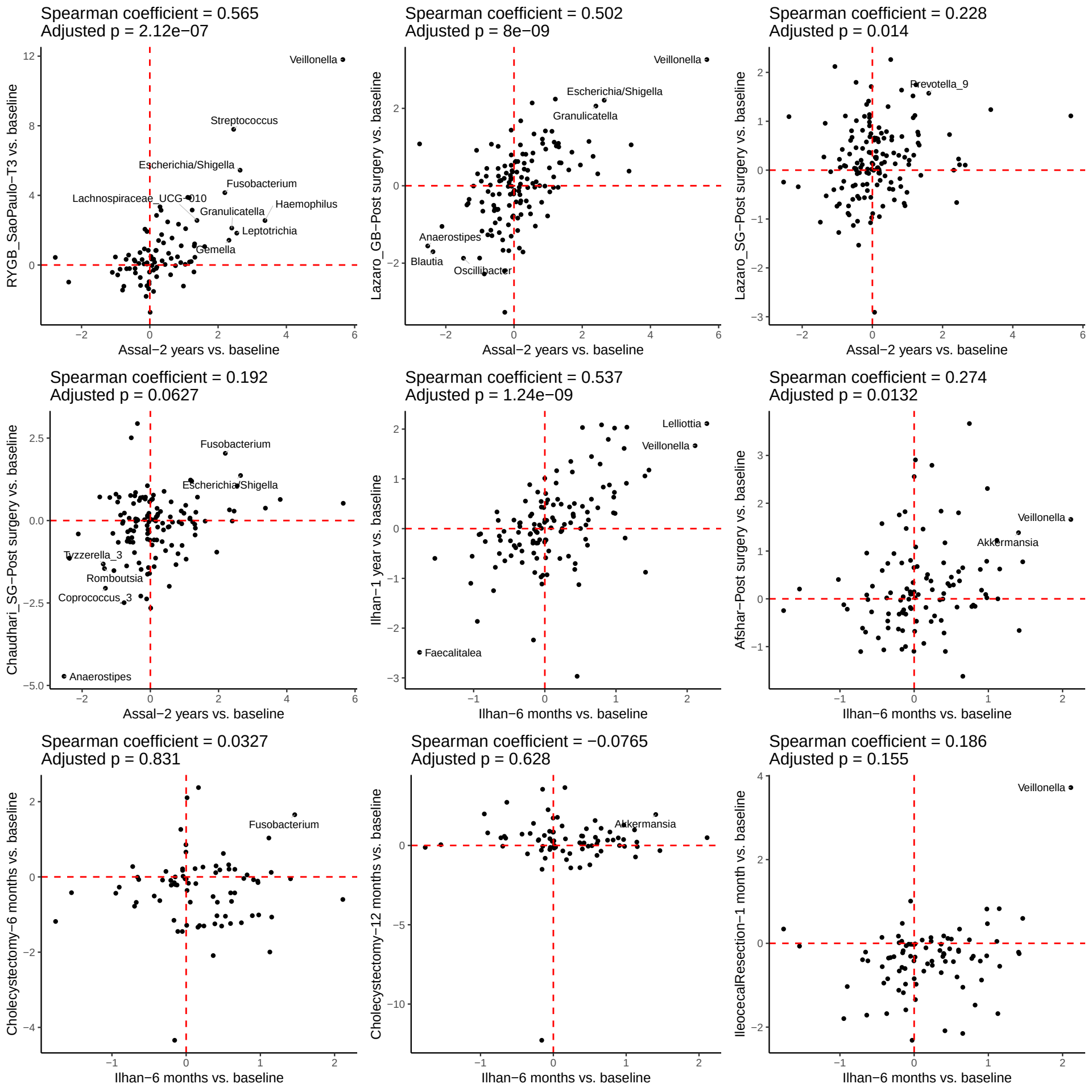

Spearman coefficient = 0.239
Adjusted p = 0.0197
Chaudhari_SG−Post surgery vs. baseline
BS−6 months vs. baseline
Fusobacterium
Spearman coefficient = 0.622
Adjusted p < 2.2e−16
Assal−1 year vs. baseline
Assal−3 months vs. baseline
Veillonella
Butyricimonas
Desulfovibrio
Mogibacterium
Streptococcus
Fusobacterium
Haemophilus
Neisseria
Lachnospiraceae_UCG−001
Akkermansia
Escherichia/Shigella
Stomatobaculum
Butyricicoccus
Turicibacter
Agathobacter
Faecalitalea
Anaerostipes
Romboutsia
Spearman coefficient = 0.65
Adjusted p < 2.2e−16
Assal−2 years vs. baseline
Assal−3 months vs. baseline
Veillonella
Haemophilus
Neisseria
Actinobacillus
Escherichia/Shigella
Anaeroglobus
Granulicatella
Streptococcus
Solobacterium
Leptotrichia
Gemella
Selenomonas
Fusobacterium
Oribacterium
Desulfovibrio
Butyricimonas
Romboutsia
Tyzzerella_3
Faecalitalea
Anaerostipes
Succinivibrio
Spearman coefficient = 0.452
Adjusted p = 2.27e−06
Ilhan−6 months vs. baseline
Assal−3 months vs. baseline
Veillonella
Fusobacterium
Akkermansia
Faecalitalea
Spearman coefficient = 0.352
Adjusted p = 0.000377
Ilhan−1 year vs. baseline
Assal−3 months vs. baseline
Alistipes
Escherichia/Shigella
Veillonella
Haemophilus
Faecalitalea
Spearman coefficient = 0.517
Adjusted p = 2.71e−08
Afshar−Post surgery vs. baseline
Assal−3 months vs. baseline
Streptococcus
Actinomyces
Rothia
Granulicatella
Gemella
Veillonella
Anaerotruncus
Atopobium
Akkermansia
Faecalibacterium
Tyzzerella_3
Spearman coefficient = −0.121
Adjusted p = 0.374
Cholecystectomy−6 months vs. baseline
Assal−3 months vs. baseline
Fusobacterium
Spearman coefficient = −0.067
Adjusted p = 0.629
Cholecystectomy−12 months vs. baseline
Assal−3 months vs. baseline
Akkermansia
Desulfovibrio
Spearman coefficient = 0.289
Adjusted p = 0.0132
IleocecalResection−1 month vs. baseline
Assal−3 months vs. baseline
Veillonella
Klebsiella
Anaeroglobus

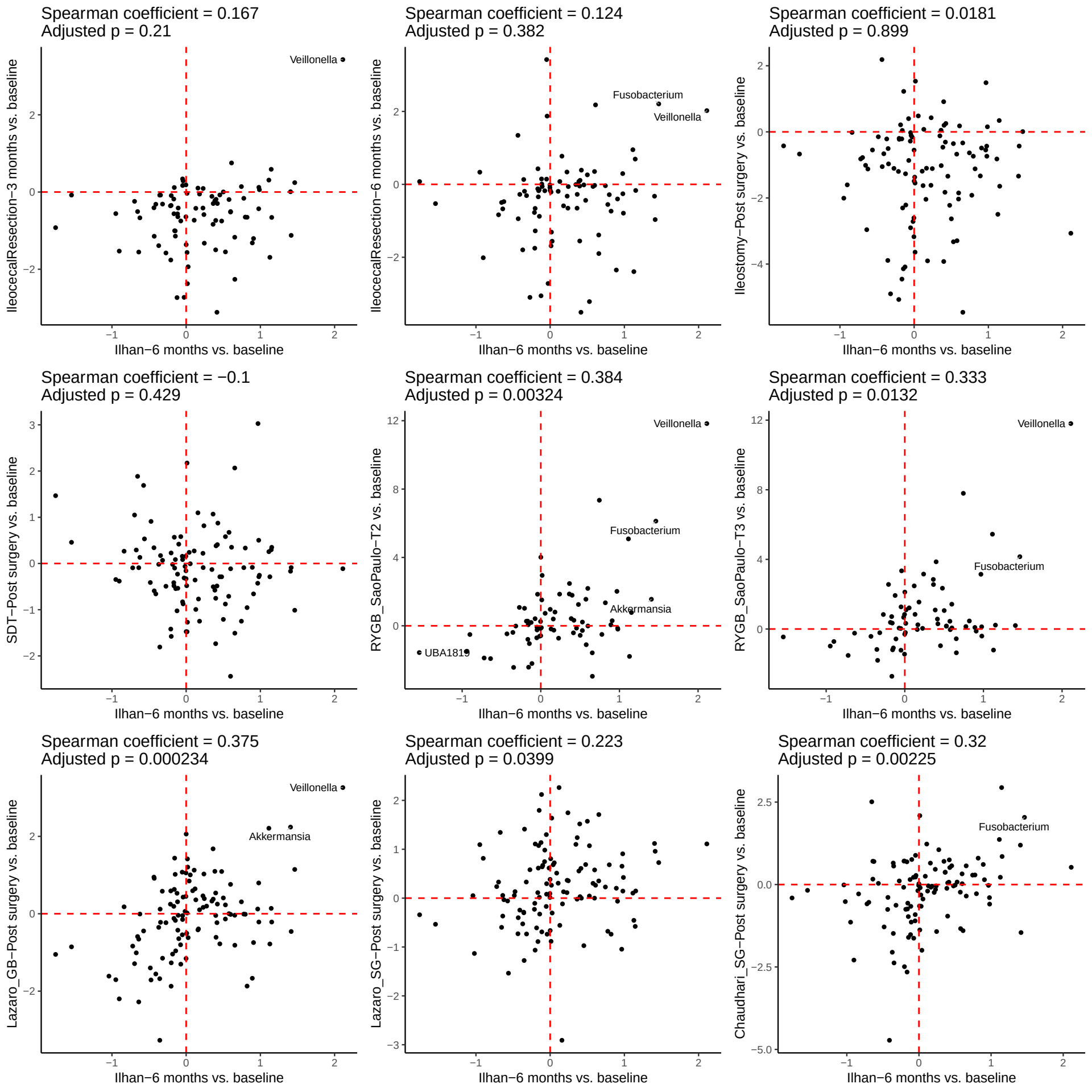

Spearman coefficient = 0.274
Adjusted p = 0.019
Veillonella
Anaeroglobus
Coprococcus_1
Faecalibacterium
IleocecalResection−3 months vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.273
Adjusted p = 0.0192
Anaeroglobus
Veillonella
Fusobacterium
Atopobium
Coprococcus_1
Faecalibacterium
IleocecalResection−6 months vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.188
Adjusted p = 0.0535
Anaeroglobus
Atopobium
Actinomyces
Tyzzerella_3
Coprococcus_1
Butyricicoccus
Faecalibacterium
Ileostomy−Post surgery vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = −0.0574
Adjusted p = 0.613
Anaeroglobus
Coprococcus_1
Faecalibacterium
SDT−Post surgery vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.568
Adjusted p = 1.88e−07
Veillonella
Streptococcus
Fusobacterium
Escherichia/Shigella
Granulicatella
Actinomyces
Klebsiella
Haemophilus
Anaerotruncus
Rothia
Gemella
Akkermansia
Fusicatenibacter
Faecalibacterium
RYGB_SaoPaulo−T2 vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.517
Adjusted p = 2.76e−06
Veillonella
Streptococcus
Escherichia/Shigella
Fusobacterium
Klebsiella
Rothia
Haemophilus
Anaerotruncus
Granulicatella
Leptotrichia
Gemella
Faecalibacterium
Fusicatenibacter
RYGB_SaoPaulo−T3 vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.509
Adjusted p = 4.21e−09
Veillonella
Akkermansia
Campylobacter
Escherichia/Shigella
Anaerotruncus
Granulicatella
Klebsiella
Actinomyces
Turicibacter
Anaerostipes
Lazaro_GB−Post surgery vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.0312
Adjusted p = 0.794
Rothia
Lazaro_SG−Post surgery vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.378
Adjusted p = 6.67e−05
Hungatella
Fusobacterium
Escherichia/Shigella
Tyzzerella_3
Coprococcus_1
Agathobacter
Romboutsia
Fusicatenibacter
Anaerostipes
Chaudhari_SG−Post surgery vs. baseline
Assal−3 months vs. baseline

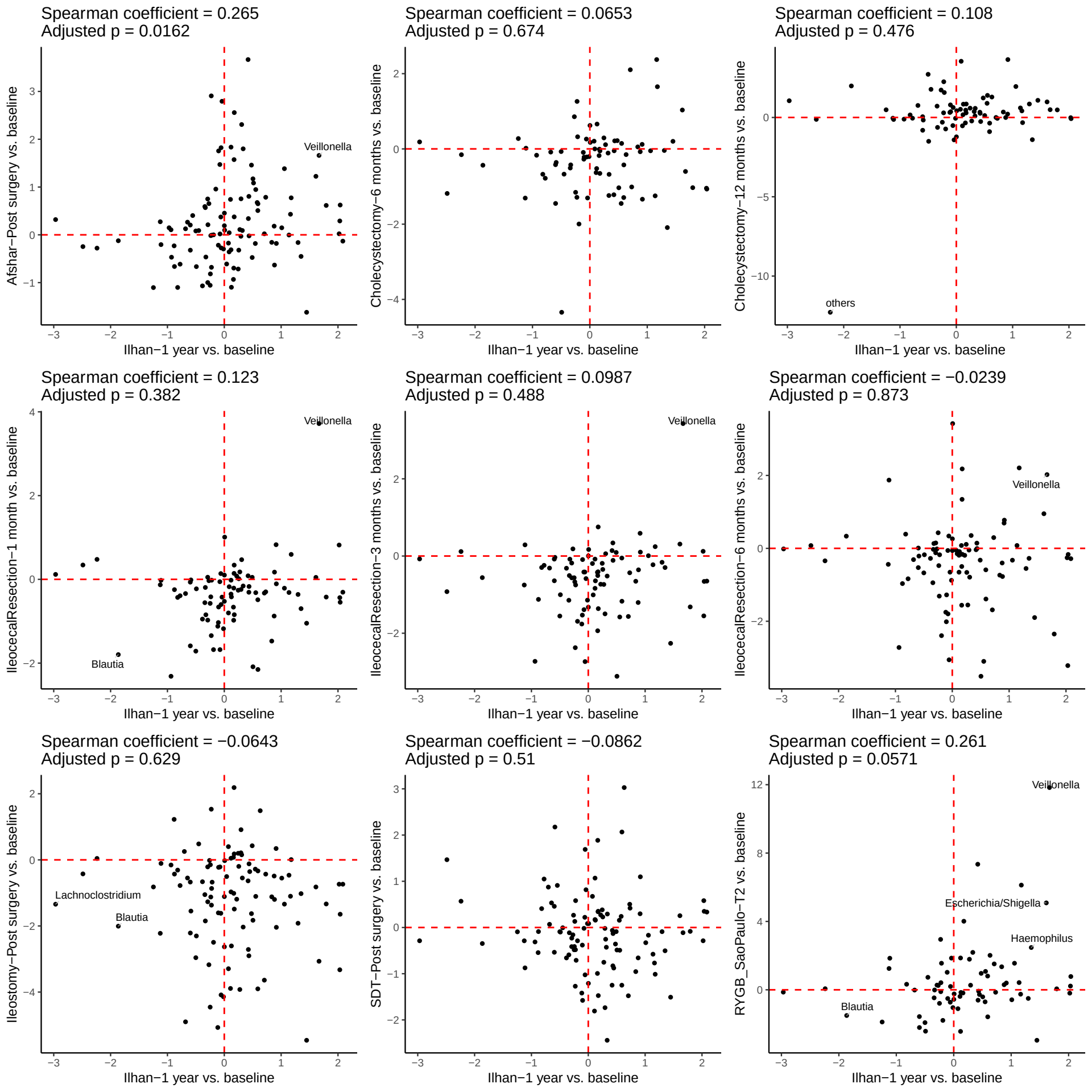

Spearman coefficient = 0.718
Adjusted p < 2.2e−16
Assal−2 years vs. baseline
Assal−1 year vs. baseline
Veillonella
Neisseria
Haemophilus
Escherichia/Shigella
Pyramidobacter
Streptococcus
Fusobacterium
Desulfovibrio
Hydrogenoanaerobacterium
Butyricimonas
Romboutsia
Oscillibacter
Faecalitalea
Blautia
Anaerostipes
Spearman coefficient = 0.467
Adjusted p = 9.57e−07
Ilhan−6 months vs. baseline
Assal−1 year vs. baseline
Veillonella
Fusobacterium
Akkermansia
Faecalitalea
Spearman coefficient = 0.411
Adjusted p = 2.18e−05
Ilhan−1 year vs. baseline
Assal−1 year vs. baseline
Phascolarctobacterium
Escherichia/Shigella
Veillonella
Haemophilus
Blautia
Faecalitalea
Spearman coefficient = 0.354
Adjusted p = 0.00032
Afshar−Post surgery vs. baseline
Assal−1 year vs. baseline
Streptococcus
Akkermansia
Veillonella
Spearman coefficient = 0.0764
Adjusted p = 0.594
Cholecystectomy−6 months vs. baseline
Assal−1 year vs. baseline
Elusimicrobium
Coprococcus_2
Fusobacterium
Spearman coefficient = −0.00661
Adjusted p = 0.967
Cholecystectomy−12 months vs. baseline
Assal−1 year vs. baseline
Akkermansia
Desulfovibrio
Family_XIII_UCG−001_group
Spearman coefficient = 0.287
Adjusted p = 0.0137
IleocecalResection−1 month vs. baseline
Assal−1 year vs. baseline
Veillonella
Blautia
Spearman coefficient = 0.176
Adjusted p = 0.157
IleocecalResection−3 months vs. baseline
Assal−1 year vs. baseline
Veillonella
Spearman coefficient = 0.184
Adjusted p = 0.139
IleocecalResection−6 months vs. baseline
Assal−1 year vs. baseline
Fusobacterium
Veillonella

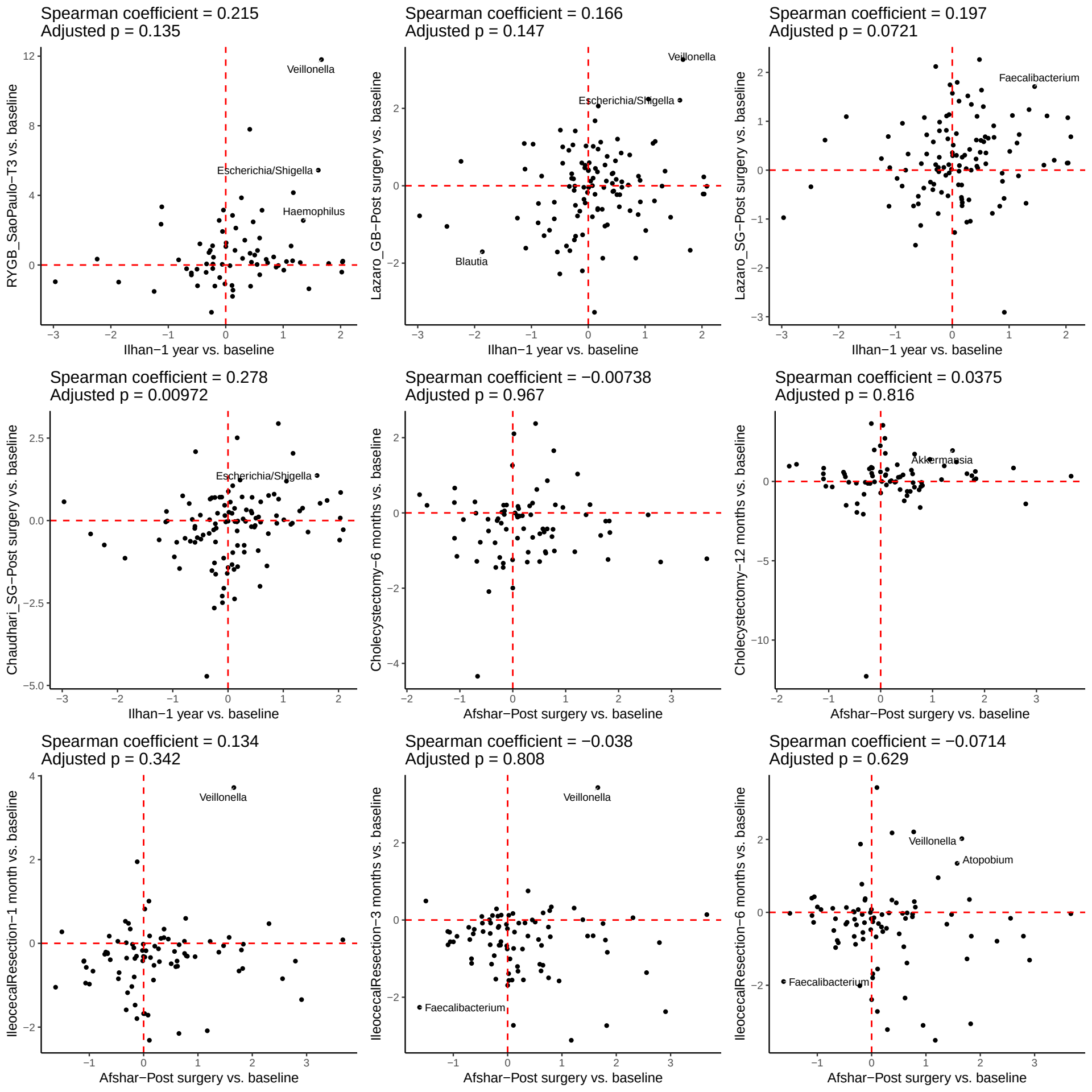

Spearman coefficient = −0.0144
Adjusted p = 0.903
Ileostomy−Post surgery vs. baseline
Assal−1 year vs. baseline
Methanobrevibacter
Pyramidobacter
Blautia
Butyricicoccus
Spearman coefficient = −0.123
Adjusted p = 0.228
SDT−Post surgery vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = 0.478
Adjusted p = 1.98e−05
RYGB_SaoPaulo−T2 vs. baseline
Assal−1 year vs. baseline
Veillonella
Streptococcus
Fusobacterium
Escherichia/Shigella
Haemophilus
Methanobrevibacter
Ruminococcaceae_UCG−003
Akkermansia
Blautia
Spearman coefficient = 0.549
Adjusted p = 5.43e−07
RYGB_SaoPaulo−T3 vs. baseline
Assal−1 year vs. baseline
Veillonella
Streptococcus
Escherichia/Shigella
Fusobacterium
Haemophilus
Ruminococcaceae_UCG−003
Catenibacterium
Methanobrevibacter
Spearman coefficient = 0.511
Adjusted p = 3.39e−09
Lazaro_GB−Post surgery vs. baseline
Assal−1 year vs. baseline
Veillonella
Akkermansia
Escherichia/Shigella
Turicibacter
Anaerostipes
Blautia
Oscillibacter
Spearman coefficient = 0.291
Adjusted p = 0.00122
Lazaro_SG−Post surgery vs. baseline
Assal−1 year vs. baseline
Ruminococcaceae_UCG−003
Spearman coefficient = 0.204
Adjusted p = 0.0471
Chaudhari_SG−Post surgery vs. baseline
Assal−1 year vs. baseline
Fusobacterium
Escherichia/Shigella
Romboutsia
Agathobacter
Anaerostipes
Spearman coefficient = 0.389
Adjusted p = 6.67e−05
Ilhan−6 months vs. baseline
Assal−2 years vs. baseline
Veillonella
Fusobacterium
Faecalitalea
Spearman coefficient = 0.325
Adjusted p = 0.00116
Ilhan−1 year vs. baseline
Assal−2 years vs. baseline
Escherichia/Shigella
Veillonella
Haemophilus
Blautia
Faecalitalea

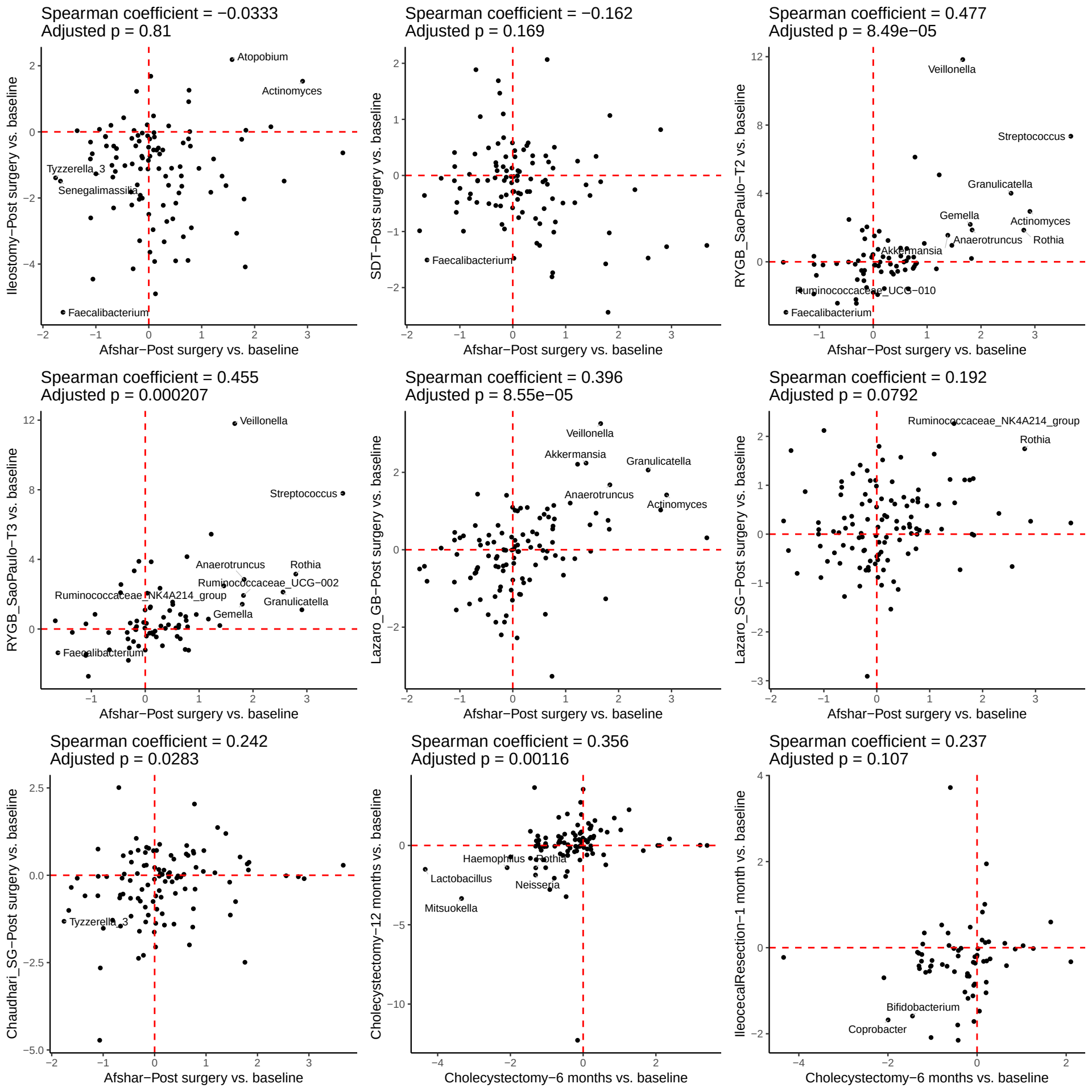

Spearman coefficient = 0.336
Adjusted p = 0.000725
Streptococcus
Granulicatella
Gemella
Veillonella
Tyzzerella_3
Afshar−Post surgery vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = −0.0304
Adjusted p = 0.831
Fusobacterium
Cholecystectomy−6 months vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = −0.0799
Adjusted p = 0.572
Desulfovibrio
Shuttleworthia
Cholecystectomy−12 months vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = 0.323
Adjusted p = 0.00488
Veillonella
Anaeroglobus
Coprococcus_3
Blautia
IleocecalResection−1 month vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = 0.248
Adjusted p = 0.0363
Veillonella
Anaeroglobus
Coprococcus_3
IleocecalResection−3 months vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = 0.282
Adjusted p = 0.0156
Anaeroglobus
Fusobacterium
Veillonella
Coprococcus_3
IleocecalResection−6 months vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = 0.0042
Adjusted p = 0.971
Anaeroglobus
Pyramidobacter
Tyzzerella_3
Blautia
Ileostomy−Post surgery vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = −0.06
Adjusted p = 0.597
Anaeroglobus
SDT−Post surgery vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = 0.451
Adjusted p = 6.67e−05
Veillonella
Streptococcus
Fusobacterium
Escherichia/Shigella
Granulicatella
Haemophilus
Gemella
Shuttleworthia
Blautia
RYGB_SaoPaulo−T2 vs. baseline
Assal−2 years vs. baseline

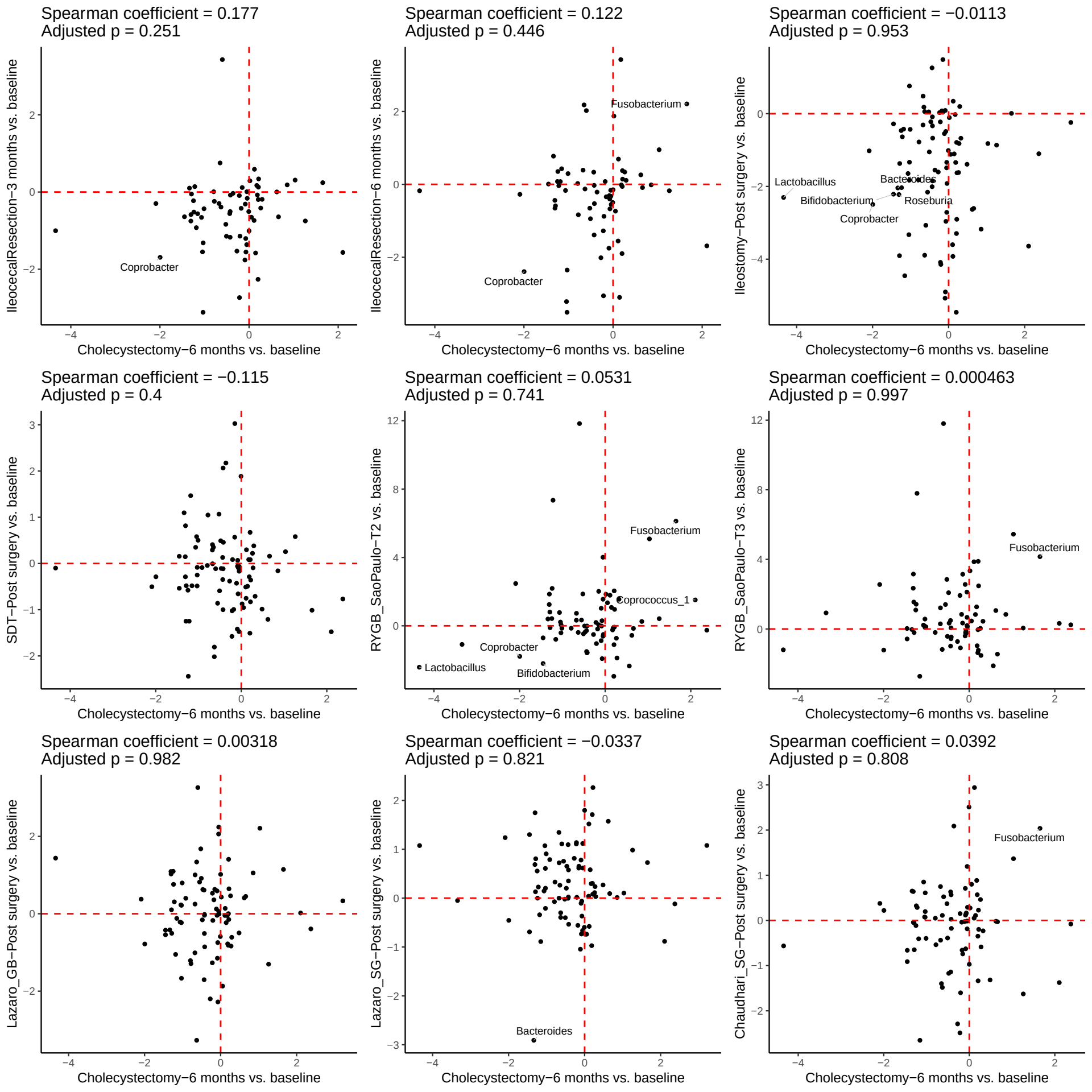

Spearman coefficient = 0.565
Adjusted p = 2.12e−07
RYGB_SaoPaulo−T3 vs. baseline
Assal−2 years vs. baseline
Veillonella
Streptococcus
Escherichia/Shigella
Fusobacterium
Lachnospiraceae_UCG−010
Granulicatella
Haemophilus
Leptotrichia
Gemella
Spearman coefficient = 0.502
Adjusted p = 8e−09
Lazaro_GB−Post surgery vs. baseline
Assal−2 years vs. baseline
Veillonella
Escherichia/Shigella
Granulicatella
Anaerostipes
Blautia
Oscillibacter
Spearman coefficient = 0.228
Adjusted p = 0.014
Lazaro_SG−Post surgery vs. baseline
Assal−2 years vs. baseline
Prevotella_9
Spearman coefficient = 0.192
Adjusted p = 0.0627
Chaudhari_SG−Post surgery vs. baseline
Assal−2 years vs. baseline
Fusobacterium
Escherichia/Shigella
Tyzzerella_3
Romboutsia
Coprococcus_3
Anaerostipes
Spearman coefficient = 0.537
Adjusted p = 1.24e−09
Ilhan−1 year vs. baseline
Ilhan−6 months vs. baseline
Lelliottia
Veillonella
Faecalitalea
Spearman coefficient = 0.274
Adjusted p = 0.0132
Afshar−Post surgery vs. baseline
Ilhan−6 months vs. baseline
Veillonella
Akkermansia
Spearman coefficient = 0.0327
Adjusted p = 0.831
Cholecystectomy−6 months vs. baseline
Ilhan−6 months vs. baseline
Fusobacterium
Spearman coefficient = −0.0765
Adjusted p = 0.628
Cholecystectomy−12 months vs. baseline
Ilhan−6 months vs. baseline
Akkermansia
Spearman coefficient = 0.186
Adjusted p = 0.155
IleocecalResection−1 month vs. baseline
Ilhan−6 months vs. baseline
Veillonella

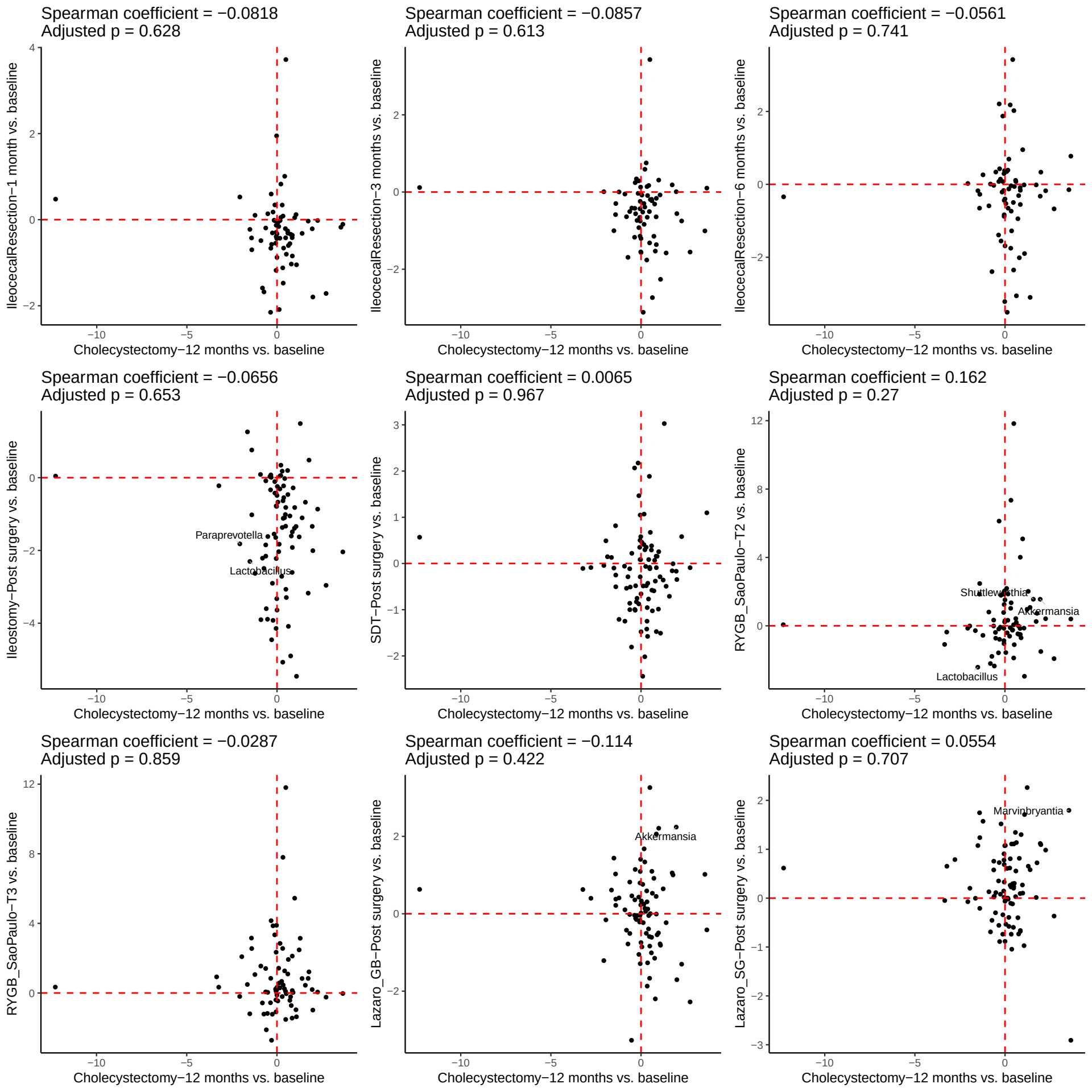

Spearman coefficient = 0.167
Adjusted p = 0.21
Veillonella
IleocecalResection−3 months vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.124
Adjusted p = 0.382
Fusobacterium
Veillonella
IleocecalResection−6 months vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.0181
Adjusted p = 0.899
Ileostomy−Post surgery vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = −0.1
Adjusted p = 0.429
SDT−Post surgery vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.384
Adjusted p = 0.00324
Veillonella
Fusobacterium
Akkermansia
UBA1819
RYGB_SaoPaulo−T2 vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.333
Adjusted p = 0.0132
Veillonella
Fusobacterium
RYGB_SaoPaulo−T3 vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.375
Adjusted p = 0.000234
Veillonella
Akkermansia
Lazaro_GB−Post surgery vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.223
Adjusted p = 0.0399
Lazaro_SG−Post surgery vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.32
Adjusted p = 0.00225
Fusobacterium
Chaudhari_SG−Post surgery vs. baseline
Ilhan−6 months vs. baseline

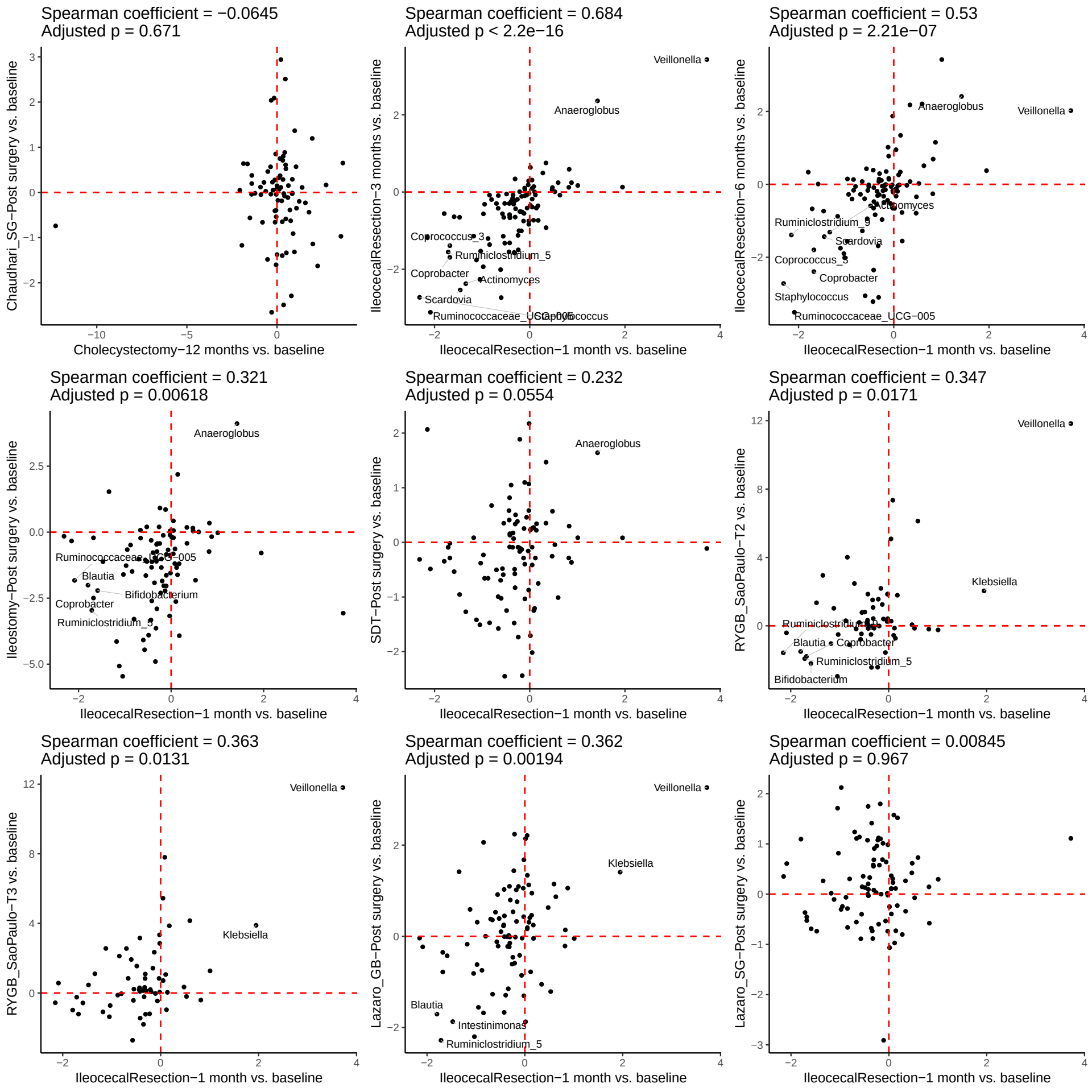

Spearman coefficient = 0.265
Adjusted p = 0.0162
Afshar−Post surgery vs. baseline
Veillonella
Ilhan−1 year vs. baseline
Spearman coefficient = 0.0653
Adjusted p = 0.674
Cholecystectomy−6 months vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.108
Adjusted p = 0.476
Cholecystectomy−12 months vs. baseline
others
Ilhan−1 year vs. baseline
Spearman coefficient = 0.123
Adjusted p = 0.382
IleocecalResection−1 month vs. baseline
Veillonella
Blautia
Ilhan−1 year vs. baseline
Spearman coefficient = 0.0987
Adjusted p = 0.488
IleocecalResection−3 months vs. baseline
Veillonella
Ilhan−1 year vs. baseline
Spearman coefficient = −0.0239
Adjusted p = 0.873
IleocecalResection−6 months vs. baseline
Veillonella
Ilhan−1 year vs. baseline
Spearman coefficient = −0.0643
Adjusted p = 0.629
Ileostomy−Post surgery vs. baseline
Lachnoclostridium
Blautia
Ilhan−1 year vs. baseline
Spearman coefficient = −0.0862
Adjusted p = 0.51
SDT−Post surgery vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.261
Adjusted p = 0.0571
RYGB_SaoPaulo−T2 vs. baseline
Veillonella
Escherichia/Shigella
Haemophilus
Blautia
Ilhan−1 year vs. baseline

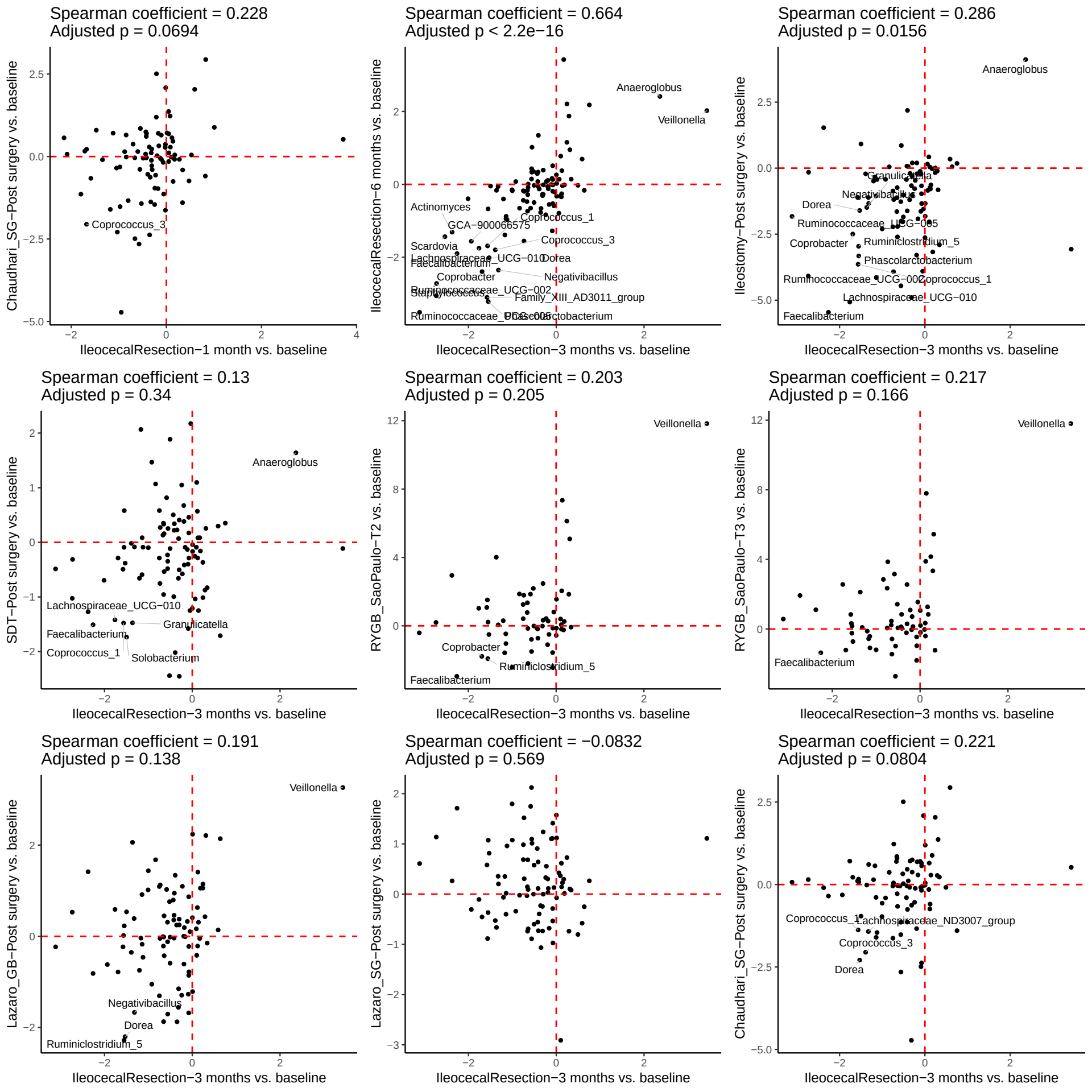

Spearman coefficient = 0.215
Adjusted p = 0.135
Veillonella
Escherichia/Shigella
Haemophilus
RYGB_SaoPaulo−T3 vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.166
Adjusted p = 0.147
Veillonella
Escherichia/Shigella
Blautia
Lazaro_GB−Post surgery vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.197
Adjusted p = 0.0721
Faecalibacterium
Lazaro_SG−Post surgery vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.278
Adjusted p = 0.00972
Escherichia/Shigella
Chaudhari_SG−Post surgery vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = −0.00738
Adjusted p = 0.967
Cholecystectomy−6 months vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = 0.0375
Adjusted p = 0.816
Akkermansia
Cholecystectomy−12 months vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = 0.134
Adjusted p = 0.342
Veillonella
IleocecalResection−1 month vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = −0.038
Adjusted p = 0.808
Veillonella
Faecalibacterium
IleocecalResection−3 months vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = −0.0714
Adjusted p = 0.629
Veillonella
Atopobium
Faecalibacterium
IleocecalResection−6 months vs. baseline
Afshar−Post surgery vs. baseline

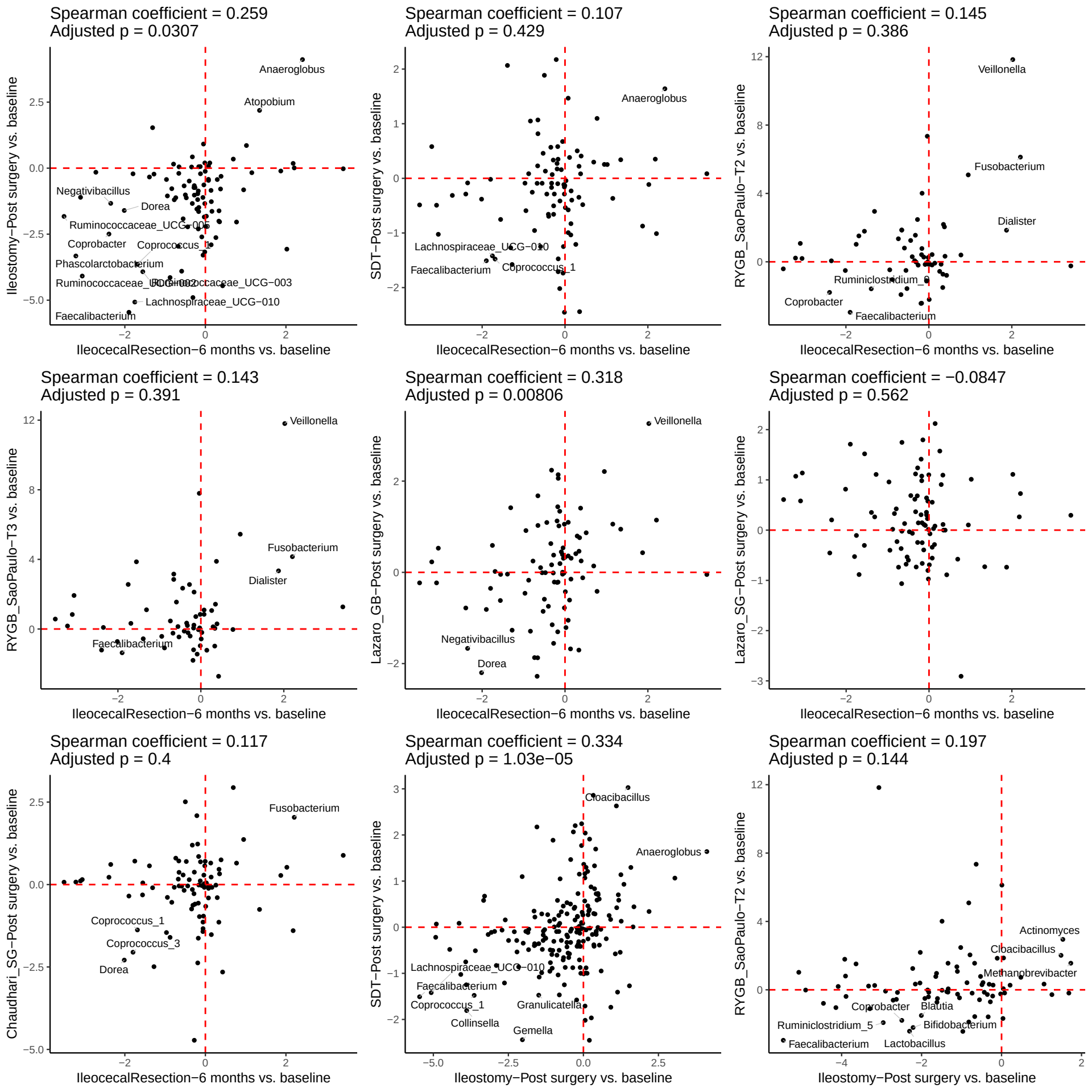

Spearman coefficient = −0.0333
Adjusted p = 0.81
Ileostomy−Post surgery vs. baseline
Afshar−Post surgery vs. baseline
Atopobium
Actinomyces
Tyzzerella_3
Senegalimassilia
Faecalibacterium
Spearman coefficient = −0.162
Adjusted p = 0.169
SDT−Post surgery vs. baseline
Afshar−Post surgery vs. baseline
Faecalibacterium
Spearman coefficient = 0.477
Adjusted p = 8.49e−05
RYGB_SaoPaulo−T2 vs. baseline
Afshar−Post surgery vs. baseline
Veillonella
Streptococcus
Granulicatella
Gemella
Actinomyces
Anaerotruncus
Rothia
Akkermansia
Ruminococcaceae_UCG−010
Faecalibacterium
Spearman coefficient = 0.455
Adjusted p = 0.000207
RYGB_SaoPaulo−T3 vs. baseline
Afshar−Post surgery vs. baseline
Veillonella
Streptococcus
Anaerotruncus
Rothia
Ruminococcaceae_UCG−002
Ruminococcaceae_NK4A214_group
Granulicatella
Gemella
Faecalibacterium
Spearman coefficient = 0.396
Adjusted p = 8.55e−05
Lazaro_GB−Post surgery vs. baseline
Afshar−Post surgery vs. baseline
Veillonella
Akkermansia
Granulicatella
Anaerotruncus
Actinomyces
Spearman coefficient = 0.192
Adjusted p = 0.0792
Lazaro_SG−Post surgery vs. baseline
Afshar−Post surgery vs. baseline
Ruminococcaceae_NK4A214_group
Rothia
Spearman coefficient = 0.242
Adjusted p = 0.0283
Chaudhari_SG−Post surgery vs. baseline
Afshar−Post surgery vs. baseline
Tyzzerella_3
Spearman coefficient = 0.356
Adjusted p = 0.00116
Cholecystectomy−12 months vs. baseline
Cholecystectomy−6 months vs. baseline
Haemophilus
Rothia
Lactobacillus
Neisseria
Mitsuokella
Spearman coefficient = 0.237
Adjusted p = 0.107
IleocecalResection−1 month vs. baseline
Cholecystectomy−6 months vs. baseline
Bifidobacterium
Coprobacter

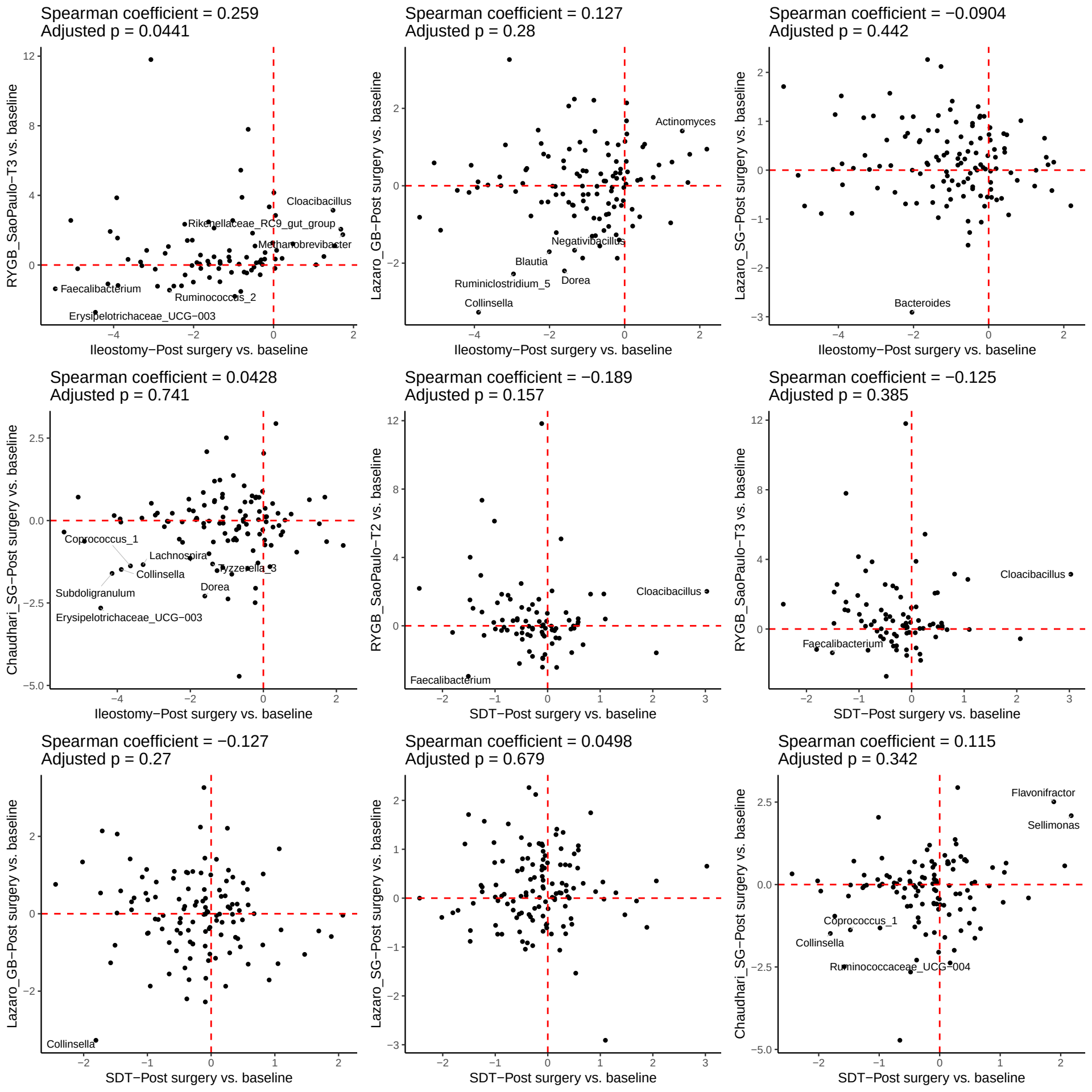

Spearman coefficient = 0.177
Adjusted p = 0.251
IleocecalResection−3 months vs. baseline
Cholecystectomy−6 months vs. baseline
Coprobacter
Spearman coefficient = 0.122
Adjusted p = 0.446
IleocecalResection−6 months vs. baseline
Cholecystectomy−6 months vs. baseline
Fusobacterium
Coprobacter
Spearman coefficient = −0.0113
Adjusted p = 0.953
Ileostomy−Post surgery vs. baseline
Cholecystectomy−6 months vs. baseline
Lactobacillus
Bacteroides
Bifidobacterium
Roseburia
Coprobacter
Spearman coefficient = −0.115
Adjusted p = 0.4
SDT−Post surgery vs. baseline
Cholecystectomy−6 months vs. baseline
Spearman coefficient = 0.0531
Adjusted p = 0.741
RYGB_SaoPaulo−T2 vs. baseline
Cholecystectomy−6 months vs. baseline
Fusobacterium
Coprococcus_1
Coprobacter
Lactobacillus
Bifidobacterium
Spearman coefficient = 0.000463
Adjusted p = 0.997
RYGB_SaoPaulo−T3 vs. baseline
Cholecystectomy−6 months vs. baseline
Fusobacterium
Spearman coefficient = 0.00318
Adjusted p = 0.982
Lazaro_GB−Post surgery vs. baseline
Cholecystectomy−6 months vs. baseline
Spearman coefficient = −0.0337
Adjusted p = 0.821
Lazaro_SG−Post surgery vs. baseline
Cholecystectomy−6 months vs. baseline
Bacteroides
Spearman coefficient = 0.0392
Adjusted p = 0.808
Chaudhari_SG−Post surgery vs. baseline
Cholecystectomy−6 months vs. baseline
Fusobacterium

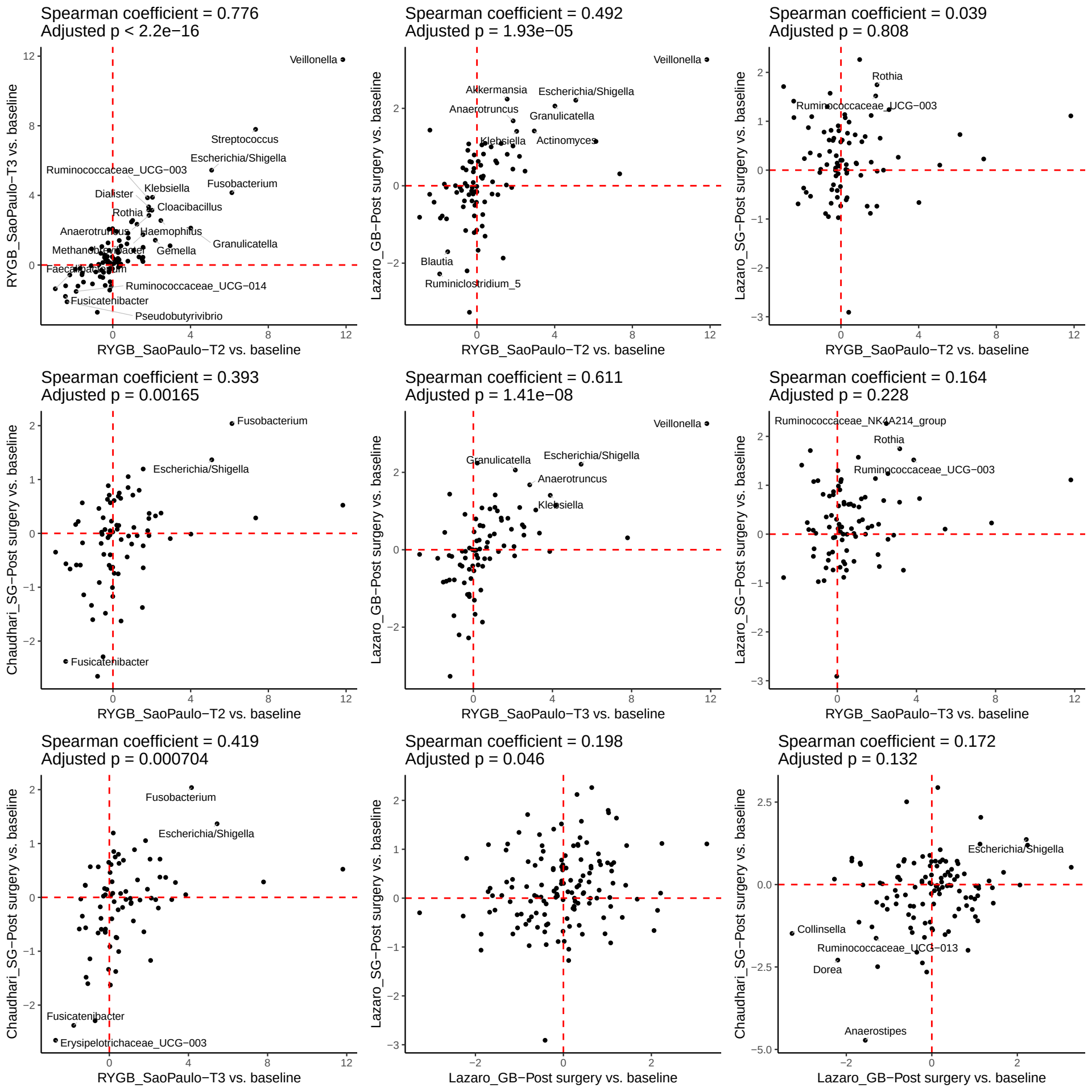

Spearman coefficient = −0.0818
Adjusted p = 0.628
IleocecalResection−1 month vs. baseline
Cholecystectomy−12 months vs. baseline
Spearman coefficient = −0.0857
Adjusted p = 0.613
IleocecalResection−3 months vs. baseline
Cholecystectomy−12 months vs. baseline
Spearman coefficient = −0.0561
Adjusted p = 0.741
IleocecalResection−6 months vs. baseline
Cholecystectomy−12 months vs. baseline
Spearman coefficient = −0.0656
Adjusted p = 0.653
Ileostomy−Post surgery vs. baseline
Paraprevotella
Lactobacillus
Cholecystectomy−12 months vs. baseline
Spearman coefficient = 0.0065
Adjusted p = 0.967
SDT−Post surgery vs. baseline
Cholecystectomy−12 months vs. baseline
Spearman coefficient = 0.162
Adjusted p = 0.27
RYGB_SaoPaulo−T2 vs. baseline
Shuttleworthia
Akkermansia
Lactobacillus
Cholecystectomy−12 months vs. baseline
Spearman coefficient = −0.0287
Adjusted p = 0.859
RYGB_SaoPaulo−T3 vs. baseline
Cholecystectomy−12 months vs. baseline
Spearman coefficient = −0.114
Adjusted p = 0.422
Lazaro_GB−Post surgery vs. baseline
Akkermansia
Cholecystectomy−12 months vs. baseline
Spearman coefficient = 0.0554
Adjusted p = 0.707
Lazaro_SG−Post surgery vs. baseline
Marvinbryantia
Cholecystectomy−12 months vs. baseline

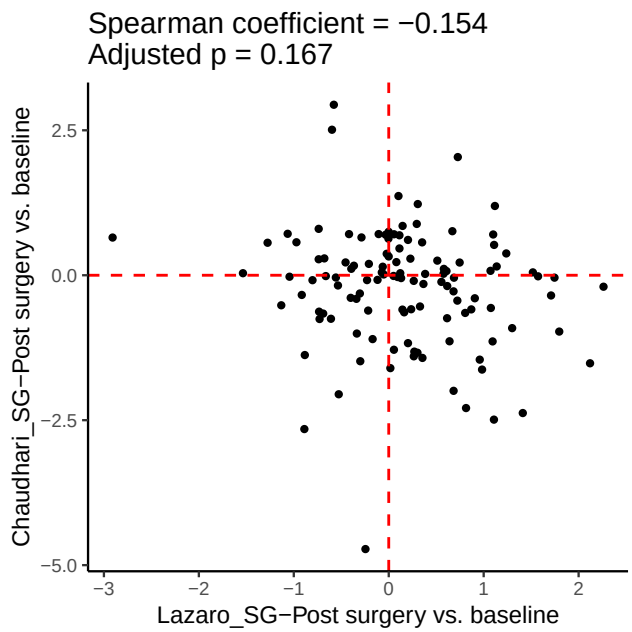

Spearman coefficient = −0.0645
Adjusted p = 0.671
Chaudhari_SG−Post surgery vs. baseline
Cholecystectomy−12 months vs. baseline
Spearman coefficient = 0.684
Adjusted p < 2.2e−16
Ileocecalresection−3 months vs. baseline
Ileocecalresection−1 month vs. baseline
Veillonella
Anaeroglobus
Coprococcus_3
Ruminiclostridium_5
Coprobacter
Actinomyces
Scardovia
Ruminococcaceae_UCG−005
Staphylococcus
Spearman coefficient = 0.53
Adjusted p = 2.21e−07
Ileocecalresection−6 months vs. baseline
Ileocecalresection−1 month vs. baseline
Anaeroglobus
Veillonella
Actinomyces
Ruminiclostridium_5
Scardovia
Coprococcus_3
Coprobacter
Staphylococcus
Ruminococcaceae_UCG−005
Spearman coefficient = 0.321
Adjusted p = 0.00618
Ileostomy−Post surgery vs. baseline
Ileocecalresection−1 month vs. baseline
Anaeroglobus
Ruminococcaceae_UCG−005
Blautia
Bifidobacterium
Coprobacter
Ruminiclostridium_5
Spearman coefficient = 0.232
Adjusted p = 0.0554
SDT−Post surgery vs. baseline
Ileocecalresection−1 month vs. baseline
Anaeroglobus
Spearman coefficient = 0.347
Adjusted p = 0.0171
RYGB_SaoPaulo−T2 vs. baseline
Ileocecalresection−1 month vs. baseline
Veillonella
Klebsiella
Ruminiclostridium_5
Blautia
Coprobacter
Ruminiclostridium_5
Bifidobacterium
Spearman coefficient = 0.363
Adjusted p = 0.0131
RYGB_SaoPaulo−T3 vs. baseline
Ileocecalresection−1 month vs. baseline
Veillonella
Klebsiella
Spearman coefficient = 0.362
Adjusted p = 0.00194
Lazaro_GB−Post surgery vs. baseline
Ileocecalresection−1 month vs. baseline
Veillonella
Klebsiella
Blautia
Intestinimonas
Ruminiclostridium_5
Spearman coefficient = 0.00845
Adjusted p = 0.967
Lazaro_SG−Post surgery vs. baseline
Ileocecalresection−1 month vs. baseline


Spearman coefficient = 0.228
Adjusted p = 0.0694
Chaudhari_SG−Post surgery vs. baseline
IleocecalResection−1 month vs. baseline
Coprococcus_3
Spearman coefficient = 0.664
Adjusted p < 2.2e−16
IleocecalResection−6 months vs. baseline
IleocecalResection−3 months vs. baseline
Anaeroglobus
Veillonella
Actinomyces
GCA−900066575
Coprococcus_1
Coprococcus_3
Scardovia
Lachnospiraceae_UCG−010
Dorea
Faecalibacterium
Coprobacter
Negativibacillus
Ruminococcaceae_UCG−002
Family_XIII_AD3011_group
Ruminococcaceae_UCG−005
Phascolarctobacterium
Spearman coefficient = 0.286
Adjusted p = 0.0156
Ileostomy−Post surgery vs. baseline
Granulicatella
Negativibacillus
Dorea
Ruminococcaceae_UCG−005
Coprobacter
Ruminiclostridium_5
Phascolarctobacterium
Ruminococcaceae_UCG−002
Coprococcus_1
Lachnospiraceae_UCG−010
Faecalibacterium
Spearman coefficient = 0.13
Adjusted p = 0.34
SDT−Post surgery vs. baseline
Anaeroglobus
Lachnospiraceae_UCG−010
Granulicatella
Faecalibacterium
Coprococcus_1
Solobacterium
Spearman coefficient = 0.203
Adjusted p = 0.205
RYGB_SaoPaulo−T2 vs. baseline
Veillonella
Coprobacter
Ruminiclostridium_5
Faecalibacterium
Spearman coefficient = 0.217
Adjusted p = 0.166
RYGB_SaoPaulo−T3 vs. baseline
Veillonella
Faecalibacterium
Spearman coefficient = 0.191
Adjusted p = 0.138
Lazaro_GB−Post surgery vs. baseline
Veillonella
Negativibacillus
Dorea
Ruminiclostridium_5
Spearman coefficient = −0.0832
Adjusted p = 0.569
Lazaro_SG−Post surgery vs. baseline
Spearman coefficient = 0.221
Adjusted p = 0.0804
Chaudhari_SG−Post surgery vs. baseline
Coprococcus_1
Lachnospiraceae_ND3007_group
Coprococcus_3
Dorea


Spearman coefficient = 0.259
Adjusted p = 0.0307
Ileostomy−Post surgery vs. baseline
IleocecalResection−6 months vs. baseline
Anaeroglobus
Atopobium
Negativibacillus
Dorea
Ruminococcaceae_UCG−005
Coprobacter
Coprococcus_1
Phascolarctobacterium
Ruminococcaceae_UCG−002
Ruminococcaceae_UCG−003
Lachnospiraceae_UCG−010
Faecalibacterium
Spearman coefficient = 0.107
Adjusted p = 0.429
SDT−Post surgery vs. baseline
IleocecalResection−6 months vs. baseline
Anaeroglobus
Lachnospiraceae_UCG−010
Faecalibacterium
Coprococcus_1
Spearman coefficient = 0.145
Adjusted p = 0.386
RYGB_SaoPaulo−T2 vs. baseline
IleocecalResection−6 months vs. baseline
Veillonella
Fusobacterium
Dialister
Ruminiclostridium_6
Coprobacter
Faecalibacterium
Spearman coefficient = 0.143
Adjusted p = 0.391
RYGB_SaoPaulo−T3 vs. baseline
IleocecalResection−6 months vs. baseline
Veillonella
Fusobacterium
Dialister
Faecalibacterium
Spearman coefficient = 0.318
Adjusted p = 0.00806
Lazaro_GB−Post surgery vs. baseline
IleocecalResection−6 months vs. baseline
Veillonella
Negativibacillus
Dorea
Spearman coefficient = −0.0847
Adjusted p = 0.562
Lazaro_SG−Post surgery vs. baseline
IleocecalResection−6 months vs. baseline
Spearman coefficient = 0.117
Adjusted p = 0.4
Chaudhari_SG−Post surgery vs. baseline
IleocecalResection−6 months vs. baseline
Fusobacterium
Coprococcus_1
Coprococcus_3
Dorea
Spearman coefficient = 0.334
Adjusted p = 1.03e−05
SDT−Post surgery vs. baseline
Ileostomy−Post surgery vs. baseline
Cloacibacillus
Anaeroglobus
Lachnospiraceae_UCG−010
Faecalibacterium
Coprococcus_1
Granulicatella
Collinsella
Gemella
Spearman coefficient = 0.197
Adjusted p = 0.144
RYGB_SaoPaulo−T2 vs. baseline
Ileostomy−Post surgery vs. baseline
Actinomyces
Cloacibacillus
Methanobrevibacter
Coprobacter
Blautia
Ruminiclostridium_5
Bifidobacterium
Faecalibacterium
Lactobacillus


Spearman coefficient = 0.259
Adjusted p = 0.0441
RYGB_SaoPaulo−T3 vs. baseline
Ileostomy−Post surgery vs. baseline
Cloacibacillus
Rikenellaceae_RC9_gut_group
Methanobrevibacter
Faecalibacterium
Ruminococcus_2
Erysipelotrichaceae_UCG−003
Spearman coefficient = 0.127
Adjusted p = 0.28
Lazaro_GB−Post surgery vs. baseline
Ileostomy−Post surgery vs. baseline
Actinomyces
Negativibacillus
Blautia
Ruminiclostridium_5
Dorea
Collinsella
Spearman coefficient = −0.0904
Adjusted p = 0.442
Lazaro_SG−Post surgery vs. baseline
Ileostomy−Post surgery vs. baseline
Bacteroides
Spearman coefficient = 0.0428
Adjusted p = 0.741
Chaudhari_SG−Post surgery vs. baseline
Ileostomy−Post surgery vs. baseline
Coprococcus_1
Lachnospira
Collinsella
Tyzzerella_3
Dorea
Subdoligranulum
Erysipelotrichaceae_UCG−003
Spearman coefficient = −0.189
Adjusted p = 0.157
RYGB_SaoPaulo−T2 vs. baseline
SDT−Post surgery vs. baseline
Cloacibacillus
Faecalibacterium
Spearman coefficient = −0.125
Adjusted p = 0.385
RYGB_SaoPaulo−T3 vs. baseline
SDT−Post surgery vs. baseline
Cloacibacillus
Faecalibacterium
Spearman coefficient = −0.127
Adjusted p = 0.27
Lazaro_GB−Post surgery vs. baseline
SDT−Post surgery vs. baseline
Collinsella
Spearman coefficient = 0.0498
Adjusted p = 0.679
Lazaro_SG−Post surgery vs. baseline
SDT−Post surgery vs. baseline
Spearman coefficient = 0.115
Adjusted p = 0.342
Chaudhari_SG−Post surgery vs. baseline
SDT−Post surgery vs. baseline
Flavonifractor
Sellimonas
Coprococcus_1
Collinsella
Ruminococcaceae_UCG−004


Spearman coefficient = 0.776
Adjusted p < 2.2e−16
Veillonella
Streptococcus
Escherichia/Shigella
Ruminococcaceae_UCG−003
Fusobacterium
Klebsiella
Dialister
Cloacibacillus
Rothia
Anaerotruncus
Haemophilus
Granulicatella
Methanobrevibacter
Gemella
Faecalibacterium
Ruminococcaceae_UCG−014
Fusicatenibacter
Pseudobutyrivibrio
RYGB_SaoPaulo−T3 vs. baseline
RYGB_SaoPaulo−T2 vs. baseline
Spearman coefficient = 0.492
Adjusted p = 1.93e−05
Veillonella
Akkermansia
Escherichia/Shigella
Anaerotruncus
Granulicatella
Klebsiella
Actinomyces
Blautia
Ruminiclostridium_5
Lazaro_GB−Post surgery vs. baseline
RYGB_SaoPaulo−T2 vs. baseline
Spearman coefficient = 0.039
Adjusted p = 0.808
Rothia
Ruminococcaceae_UCG−003
Lazaro_SG−Post surgery vs. baseline
RYGB_SaoPaulo−T2 vs. baseline
Spearman coefficient = 0.393
Adjusted p = 0.00165
Fusobacterium
Escherichia/Shigella
Fusicatenibacter
Chaudhari_SG−Post surgery vs. baseline
RYGB_SaoPaulo−T2 vs. baseline
Spearman coefficient = 0.611
Adjusted p = 1.41e−08
Veillonella
Granulicatella
Escherichia/Shigella
Anaerotruncus
Klebsiella
Lazaro_GB−Post surgery vs. baseline
RYGB_SaoPaulo−T3 vs. baseline
Spearman coefficient = 0.164
Adjusted p = 0.228
Ruminococcaceae_NK4A214_group
Rothia
Ruminococcaceae_UCG−003
Lazaro_SG−Post surgery vs. baseline
RYGB_SaoPaulo−T3 vs. baseline
Spearman coefficient = 0.419
Adjusted p = 0.000704
Fusobacterium
Escherichia/Shigella
Fusicatenibacter
Erysipelotrichaceae_UCG−003
Chaudhari_SG−Post surgery vs. baseline
RYGB_SaoPaulo−T3 vs. baseline
Spearman coefficient = 0.198
Adjusted p = 0.046
Lazaro_SG−Post surgery vs. baseline
Lazaro_GB−Post surgery vs. baseline
Spearman coefficient = 0.172
Adjusted p = 0.132
Escherichia/Shigella
Collinsella
Ruminococcaceae_UCG−013
Dorea
Anaerostipes
Chaudhari_SG−Post surgery vs. baseline
Lazaro_GB−Post surgery vs. baseline


Spearman coefficient = −0.154
Adjusted p = 0.167
Chaudhari_SG−Post surgery vs. baseline
2.5
0.0
−2.5
−5.0
−3
−2
−1
0
1
2
Lazaro_SG−Post surgery vs. baseline